Connectivity and Local Opportunity

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Abstract

Where you grow up shapes life chances. Remoteness and connectivity are important dimensions of this spatial distribution of opportunity, particularly in a low and middle-income country setting. However, road building creates winners and losers as economic activity adjusts in spatial equilibrium. Using plausibly exogenous variation in connectivity, and a movers design, I find that growing up in a one-standard-deviation more connected location raises local opportunity, measured as primary school completion rates, by 12% across Benin, Cameroon, and Mali. To understand aggregate effects and mechanisms, I solve and estimate a quantitative spatial economics model with education choice embedded within a place-effects framework. Road building since 1970 has increased opportunity by 11.5%, but 17% of locations were negatively affected. Although easier moves to opportunity explain 83% of the overall effect, the impact of endogenous spatial adjustments decreases the impact by 6pp in some locations, and increases it by 9.3pp in others.

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1 Introduction

Where you grow up is an important determinant of later life outcomes. These differences are perhaps most stark in the context of low and middle-income countries. They are also potentially more consequential, stoking regional dissatisfaction, reducing allocative efficiency, and causing misery for those in "left behind" areas. One potential policy solution is to move people to areas of higher opportunity, but this is not scalable. Instead, in this paper, I consider the possibility of moving the spatial distribution of opportunity itself. To do this, policymakers need to know what characteristics of a location cause it to be a place of high or low opportunity.

I consider how changes in local connectivity through road building shape the geography of opportunity in Benin, Cameroon, and Mali. I define local opportunity as the causal effect of growing up in a given location on the probability of completing primary education, following (Alesina et al., 2021; Chetty and Hendren, 2018b; Milsom, 2023).[1] Individuals growing up in a location that becomes better connected will become more exposed to high-return areas as it becomes easier to move to them from a given origin location (Hsiao, 2024), or because greater connectivity may cause the underlying spatial distribution of local opportunities to shift. Connectivity, however, need not always have a positive impact; population influxes could crowd out opportunities, and changes in the spatial distribution of demand may lead to local decreases.

Recovering the causal effect of changes in connectivity through road building on local opportunity is challenging for four key reasons. First, measuring changing connectivity through road building is complex because the impact of any given road will depend on the entire pre-existing road network and distribution of economic activity (Donaldson, 2022). Second, as a result, the impact of any given change in the network will spill over in some capacity to all other locations, prohibiting the possibility of any pure control locations. Third, road building is far from random; policymakers may build roads to service growing locations or galvanise flagging ones, and any derived measure of connectivity will inherit this endogeneity. Finally, changes in local connectivity may change the selection of individuals who move across space, potentially causing primary completion to rise (or fall) in a location simply due to a change in the composition of the local population. In this paper, I am interested in the causal effect of place on primary education completion and not in this selection effect, and so require a strategy to separately identify these channels.

[1] I focus on primary education completion for three main reasons. First, measures of income commonly used in high-income countries are less appropriate in this setting, where the majority of workers are not waged, and subsistence agriculture is common. Second, primary completion is the most salient margin of education in my setting, with only 7% of my overall sample going on to complete secondary school. Third, primary completion is widely available in data sources with sufficient geographic granularity and size to be amenable to my analysis.

To overcome these challenges, I first set up a canonical place-effects model and embed this within a quantitative spatial economics model with education and location choice, expanding upon Hsiao (2024) by endogenising local real wages in spatial equilibrium. This general framework delivers a parsimonious sufficient statistic result: The impact of roads on local opportunity can be summarised by a location's market access (Donaldson and Hornbeck, 2016). To measure market access, I combine census data with comprehensive historical road maps that I digitise. These digitised maps allow me to isolate variation in market access due solely to road building and upgrading. To overcome the challenges of endogeneity and selection into location, I combine two identification strategies. First, I use a movers design following Chetty and Hendren (2018a,b); Chiovelli et al. (2025) and others to isolate place effects by using variation in exposure to locations due to differences in the age children move across locations. Second, to overcome the endogeneity of road building, I use an instrumental variables strategy leveraging variation in market access due to far-away road building Faber (2014) and develop a novel not-on-least-cost-path identification strategy which, intuitive arguments and simulation evidence show, is effective in dealing with the endogeneity of roads.

Combining these approaches with the sufficient statistic result. I find that growing up in a one standard deviation higher market access location increases the probability of completing primary school by 7 percentage points (12%) on average. This result is robust to reasonable variations in specification, only using within-family variation in exposure to locations, and controlling for endogenous exposure to exogenous shocks (Borusyak and Hull, 2023). However, this reduced-form impact does not allow for the quantification of the aggregate impact of road building since 1970, nor does it shed light on the underlying mechanisms. By leveraging the full structure of the model, I now turn to these questions.

To quantify the impact of road building since 1970 on the spatial distribution of opportunity, I then solve the model using exact-hat algebra and identify the remaining parameters from the reduced form results, and using plausibly exogenous variation following Bryan and Morten (2019). I find that road building since 1970, on average, increases local opportunity by 11.5% — but that this average hides considerable heterogeneity. 17% of locations saw negative aggregate effects, and 19% of locations saw increases in excess of 20%. These impacts suggest that road building is an effective policy lever in moving the spatial distribution of opportunity in Benin, Cameroon, and Mali, but also that location-specific heterogeneity is important. In general, road building is not a silver bullet.

Changes in the road network can alter the spatial distribution of local opportunity through two main channels. First, it may allow greater moves to opportunity as it is now easier for individuals to migrate to areas of preexisting high opportunity. Second, it may move the underlying spatial distribution of opportunity itself as labour and goods markets clear in spatial equilibrium. Using the model I can decompose average and location-specific effects into these components. I find that the direct "moving to opportunity" effect is (almost) always positive, and shutting down any endogenous adjustment generates an average impact of road building 2pp lower than in the full model. This is intuitive; falling transport costs will necessarily increase local opportunity as higher-wage locations are now cheaper to migrate to from all origins. Allowing labour markets to clear in spatial equilibrium, but fixing goods markets, causes a considerable leftward shift in the distribution of -52.7pp. Areas with the highest real wages see the largest increases in in-migration, and this labour supply shock puts downward pressure on wages. Although sending locations will see increases in wages, as only the migration-cost conditional highest wage location is relevant, opportunity will fall in all areas. Allowing the goods market to also endogenously adjust in spatial equilibrium undoes all of this negative effect, as higher-population locations also have higher demand. In sum, endogenous responses cause the impact of roads built since 1970 to be modestly higher, but this hides considerable heterogeneity across locations. The 10th percentile of the impact of endogenous responses is -5.8pp, and the 90th percentile is 9.3pp. Many locations with little direct effect have large impacts due to endogenous responses, meaning that for a sizable minority of locations, the moving distribution of opportunity, rather than moving to opportunity, is the most important force.

This paper contributes to the literature in three main ways. First, it contributes to the literature studying spatial inequality and place effects (Chetty and Hendren, 2018a,b; Deutscher, 2020; Laliberté, 2021; Van Maarseveen, 2025; Alesina et al., 2021; Rojas Ampuero, 2022; Chetty et al., 2016; Bryan et al., 2014; Chiovelli et al., 2025), which previously has focused on estimating the causal effect of place. I add to this by going a step further and asking how policy can alter the distribution of causal place effects. A recent strand of this literature has considered place effects within a general equilibrium setting (Chyn and Daruich, 2025; Eckert and Kleineberg, 2024). I build on this literature by considering changes in connectivity in a low and middle-income country setting, and developing a framework that allows greater location-heterogeneity (relative to Chyn and Daruich (2025)) and takes a market access approach with costly trade as well as migration. In this paper, I consider how policymakers could move opportunity to people via road building in Benin, Cameroon, and Mali. I find that roads are an effective policy tool to move local opportunity, but that it is not a silver bullet — affects are heterogeneous and can be negative.

Second, I contribute to the literature on quantitative spatial economic modeling by combining the canonical place-effects framework with a quantitative spatial economics model with education choice (Redding and Rossi-Hansberg, 2017; Allen et al., 2020; Donaldson and Hornbeck, 2016; Donaldson, 2018; Allen and Arkolakis, 2023). Empirically I consider the effect of roads following (among others) Graff (2024); Kebede (2024); Sotelo (2020);

Adamopoulos (2025); Castaing Gachassin (2013), and Morten and Oliveira (2024). A smaller literature considers the interaction between observed educational attainment and trade (Fujimoto et al., 2023; Khanna, 2022; Hsiao, 2024; Edmonds et al., 2010; Atkin, 2016). Most related to this work, Adukia et al. (2020), Asher and Novosad (2020), and Mukherjee (2012), who consider the impact of connecting villages in India to the main road network on educational and economic outcomes, and Hsiao (2024) who considers educational investments in spatial equilibrium in the setting of Indonesia. I build on these papers by considering the impact of large-scale inter-city road-building in a different empirical setting, focusing on local opportunity rather than observed primary completion, and allowing for education demand to respond endogenously in spatial equilibrium. I show that roads change the spatial distribution of opportunity through the direct channel of lowering the cost of moving to existing high-return areas (moving to opportunity) and by changing the underlying distribution of returns through the endogenous reaction of trade and migration.

Lastly, I contribute to the literature on identifying the impact of changes in connectivity by developing a novel identification strategy using an iterative not-on-least-cost-path approach. An established literature looks at the causal impacts of colonial railways in Sub-Saharan Africa, such as Jedwab and Moradi (2016) and Jedwab, Kerby, and Moradi (2017). This has more recently been supplemented by work looking at roads (Moneke (2020); Jedwab and Storeygard (2021); Banerjee, Duflo, and Qian (2020); Faber (2014)), and bridges (Brooks and Donovan (2020); Zant (2022)). I contribute to this literature by expanding on the existing *far-away* variation strategy and developing an alternative to the straight-line instrument, or incidental-middle approach Redding and Turner (2015); Michaels (2008) — one which can be applied in all settings that result in a market access relationship.[2]

The rest of this paper proceeds as follows: section 2 describes the setting and data, section 3 elucidates the model, section 4 explains the identification approach and presents the reduced form regression results, section 5 quantifies the impact of road building since 1970 and considers mechanisms, and finally section 6 concludes.

[2] I also speak to the related literature that has considered the more reduced-form impacts of transport infrastructure on local outcomes. The conclusion from this literature is somewhat ambiguous. Faber (2014) finds that incidentally connected periphery areas in China have lower GDP relative to similar not-connected areas. On the other hand, Baum-Snow et al. (2017) and Banerjee et al. (2020) find positive impacts of being connected in a similar setting, and Jedwab and Storeygard (2021) similarly in Sub-Saharan Africa. Baum-Snow et al. (2020) attempts to reconcile this, again considering roads in China, by showing that impacts depend on the ex-ante urban hierarchy: Core cities benefit at the expense of periphery ones. This paper can be seen as taking the implicit heterogeneity seriously by allowing the impact of every road on every location to differ, and for this to depend on the entire pre-existing road network and distribution of economic activity. This highlights the importance of not generalising from one road to another.

2 Setting & data: Benin, Cameroon, & Mali 1970-2020

Benin, Cameroon, and Mali together provide an excellent setting to study how changes in connectivity affect spatial inequality, as well as each country being an important setting in which to study these questions in and of themselves. This is because each country displays considerable past variation in local connectivity due to road building, which I will leverage when estimating the effects of changes in connectivity on spatial inequality. Additionally, due to low preexisting levels of paved road coverage (Gwilliam, 2011; Foster and Briceño-Garmendia, 2010) and high anticipated urbanisation and population growth (UN, 2018), considerable investment in road infrastructure is expected in the near future, making this a particularly direct and policy-relevant setting to study the impacts of road building.

Benin, Cameroon, and Mali also have the advantage of being the only countries in sub-Saharan Africa with data sufficiently rich to identify the impact of changes in connectivity on the causal effect of place on primary education completion. To estimate causal place effects using a movers design, it is necessary to observe where an individual is when the census was taken, their previous location, their birth location, and how long they have resided in their current location. To uncover causal place effects, previous research (Chetty and Hendren, 2018a; Laliberté, 2021; Deutscher, 2020) has mainly relied on administrative data that is not typically publicly available. This data is necessary for these studies to observe full migration histories and to match child and parent outcomes over long time frames. Such rich data is not needed to estimate place effects in the censuses I use; it's possible to discern migration histories from cross-sectional evidence. In addition to migration information, I require data on primary education completion,[3] and age, at a granular and consistent geographic level over multiple cross-sectional waves. Only three countries in Sub-Saharan Africa — Benin, Cameroon, and Mali fit the stringent data requirements.

In the remainder of this section, I briefly sketch out the historical context of road building within Benin, Cameroon, and Mali, before describing the two main data sources: Historical road maps and the census data in more detail.

[3]In Benin, Cameroon, and Mali, education is compulsory for the first six years of schooling between the ages of 6 and 11/12, which covers primary school. Benin has a 6-4-3 education system; those who state they attend Islamic school are dropped from the sample. See Appendix 10 for details. Cameroon has a dual education system with a French-speaking 6-4-3 structure and a minority English-speaking 7-5-2 structure in some parts of the country; this is mapped onto the 6-4-3 system in my data. Any such location-specific differences will be taken into account in my analysis. Mali follows a 6-3-3 system during the period I study.

2.1 Historical context

2.1.1 Benin

Road infrastructure in the territory of present-day Benin has deep antecedents in precolonial state formation. For example, the Kingdom of Dahomey maintained a formalized "Royal Road" that linked the Atlantic port of Whydah (Ouidah) to the royal capital at Abomey, supporting royal mobility and administrative authority along a core political axis (Alpern, 1999). From the late nineteenth century, road development was reshaped by French colonial rule. In this period, transport priorities reoriented toward coastal access and administrative control. After independence (1960), Benin's early postcolonial period was marked by severe political instability (multiple coups between 1963 and 1972), which constrained long-horizon public investment planning, including in transport infrastructure.

Benin's post-1970 road policy was somewhat hampered by political and macroeconomic shocks: the 1972 coup brought Mathieu Kérékou to power, followed by an explicit Marxist-Leninist turn from 1974 and the proclamation of the People's Republic of Benin in 1975. In parallel, external financing from the World Bank came online e.g., the World Bank's Highway Maintenance and Engineering Project (Bank, 1970). By the late 1980s–early 1990s, economic stress and reform pressures culminated in Benin's "National Conference" opening on 19 February 1990, widely treated as a watershed in Francophone Africa's democratic transitions and closely intertwined with adjustment-era economic liberalization. Over this period, the mounting costs of maintaining the existing infrastructure increasingly bite (World Bank, 1994).

From the 2000s, Benin's road strategy continues to shift from network expansion toward preservation of the existing network, particularly on the port–corridor system anchored by Cotonou. For example, the World Bank's Transport Infrastructure Rehabilitation and Maintenance Project explicitly "signaled a shift" toward maintenance of existing road and port assets and protecting past investments. However, this framing coexisted with large upgrading programs such as the AfDB-supported Djougou–N'dali improvement project and related works in northern Benin.

2.1.2 Cameroon

Prior to motor-road systems, mobility depended heavily on footpaths, porterage, and waterways. Recent work has traced modern road corridors in Cameroon to the colonial era, when transport infrastructure served territorial control and resource extraction (Tchindjang et al., 2005). These "main artery" corridors built during German colonial rule, from circa 1900—initially used for conquest, porterage, and forced labour logistics—later became the backbone of Cameroon's contemporary trunk road system. Cameroon's decolonization then produced a spatially and institutionally complex situation: following a UN-supervised plebiscite in February 1961, southern British Cameroons united with the newly independent Republic of Cameroon to form a federal state.

Post-1970 road policy developed alongside major constitutional and economic changes. The federal system was abandoned with the move to a unitary state in 1972, which redefined the governance environment for national trunk-road planning and maintenance responsibilities. In the same broad period, the World Bank's Second and Third Highway Projects (Bank, 1982) targeted key trunk corridors—including the Transcameroon route and the Douala–Foumban road. However, Cameroon's mid-1980s commodity-price shock and associated recession, followed by IMF-supported structural adjustment, reduced budgets to road building and complicated routine and periodic maintenance. By 1990, chronic maintenance shortfalls led to the creation of a dedicated Road Fund under Law, which became operational by the late 1990s.

2.1.3 Mali

Mali's connectivity long predates modern roads. Precolonial economic integration relied on the Niger River valley and trans-Saharan caravan routes, with Timbuktu and related nodes historically positioned as trading posts on caravan corridors and as major centers of Islamic learning and commerce. Much like in Benin and Cameroon, under colonial rule (French) the development of the transport network shifted towards military, control, and extraction priorities. After independence (1960), Mali entered a relatively volatile political period, with a military takeover in 1968 shaping state capacity and investment planning—including transport.

By the 1980s, World Bank operations explicitly framed objectives around protecting a "priority" road network culminating in projects that combined rehabilitation with reforms to the construction and maintenance sectors. Maintenance became a standalone policy object by the late 1980s: the Road Maintenance Project (Association, 1981), for example, reflects a recognition that deferred maintenance was becoming a serious problem. Politically, Mali also experienced major regime change with the 1991 overthrow of military rule and the restoration of a civilian government in 1992.

The Malian Road Fund became operational in July 2002. It created maintenance resources through axle-load charges, a fuel levy, toll fees on selected segments, and budgetary allocations. These reforms, however, unfolded under security and governance shocks: dissatisfaction with the government's handling of the northern conflict precipitated the 2012 coup, followed by sustained warfare and insurgency pressures that have repeatedly displaced policy attention and disrupted infrastructure delivery and upkeep.

2.2 Census data

I use rich and comprehensive census data to deduce variation in local opportunity. I use data on 8 million observations from 1976 to 2013, across 164 localities from every available census from Benin, Cameroon, and Mali.[4]. In the censuses described, I can geo-locate individuals at the second administrative unit level. These localities have a median population of 267,000 across all samples. Benin's Communes have a median population of 103,000 with an inter-quartile range of 71,000 to 173,000. Mali's circles have a median population of 308,000 with an inter-quartile range of 197,000 to 520,000. Cameroon's departments have a median population of 456,000 with an inter-quartile range of 225,000 to 907,000. A broadly comparable geographical unit in the US would be commuter zones. In Benin, I use censuses from 1992, 2002, and 2013, encompassing over 2.2 million individual-level observations in 77 consistent localities (Communes). In Cameroon, I use censuses from 1976, 1987, and 2005, covering over 3.4 million individual-level observations in 39 consistent localities (Departments). Finally, in Mali, I use censuses from 1998, and 2009, giving over 2.4 million individual-level observations in 48 consistent localities (Circles). In total, this gives me a sample of over 8 million individual-level observations across 164 localities and 444 locality-year cells. A benefit of using this data is that it is freely and publicly available from IPUMS International.

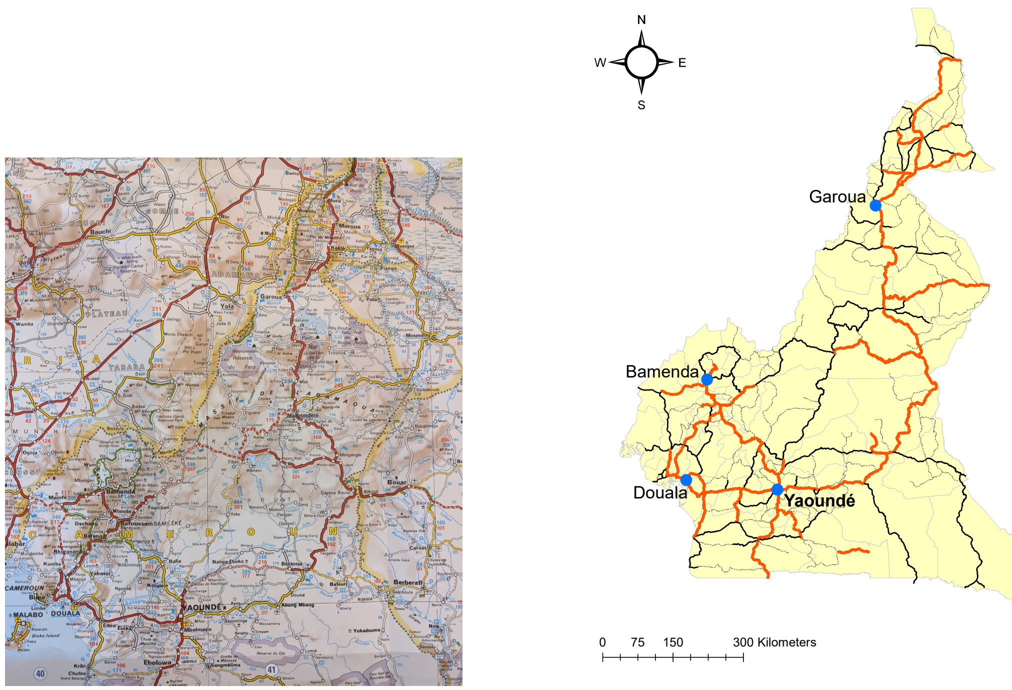
I use the causal effect of growing up in a given location on the probability of completing primary school as my main measure of local opportunity for three main reasons. First, due to the large informal sector, it is unclear whether later life income, as used in other contexts, is appropriate here. Additionally, data on primary completion rates are available at a fine geographic level over time and are less likely to suffer from significant measurement error. Second, primary completion rates are correlated with opportunity more broadly defined in later life. In this setting, individuals who have completed primary school are less likely to work in agriculture, have better housing quality, and greater returns to education (Psacharopoulos and Patrinos, 2018). Finally, primary schooling is the most salient margin of education. In my sample, about a third of individuals have completed primary school, but only 7% have completed secondary school. Thus, although defensible as a relevant and general measure of local opportunity, primary education completion cannot speak to all potential dimensions, and so opportunity in this paper should be taken to mean local *educational* opportunity.

[4]Data is accessed from IPUMS (2020) and consists of 10% samples, with thanks to the National Institute of Statistics and Economic Analysis in Benin, the Central Bureau of Census and Population Studies in Cameroon and the National Directorate of Statistics and Informatics in Mali, who provided the underlying data.

2.3 The changing geography of connectivity: Digitising historical Michelin road maps

Road data comes from historical Michelin maps, which I have digitised from the following years: 2019, 2012, 2003, 1986, 1976, and 1969. In these maps, it's possible to consistently classify roads into highways, paved roads, improved roads (laterite or gravel), and dirt roads. This classification provides a full description of the (main inter-city) roads over time since 1969 in each country. The ability to distinguish road type is of particular importance, as much of the variation in connectivity, especially in later years, comes from upgrading roads rather than building new ones. Figure 1 gives an example of this process. Panel 1a shows an image of the raw Michelin map of Cameroon in 2019, and panel 1b shows the digitised version.

Figure 1 Digitizing Michelin road maps — Cameroon 2019



(a) Michelin map (b) Digitized map

Notes: This figure shows in panel 1a the original Michelin road map of Cameroon in 2019 and in panel 1b the digitised version. In panel 1b, thick red lines are paved roads, dark black lines are improved roads, and grey lines are dirt tracks. Note that in panel 1a the colour of roads denotes their importance/ frequency of use for long-range trucking and does not necessarily reflect their size or quality, which is denoted instead by the thickness of outlines.

Michelin maps are themselves constructed using four main sources (Jedwab and Storeygard (2021)): the previous Michelin map, government road maps, local information from Michelin tyre stores across Africa, and finally direct correspondence from users. It is generally thought that this process leads to consistent and accurate road mapping over time. Indeed, the success of Michelin maps relied on them being a trustworthy source of information, and so Michelin had a vested interest in providing accurate maps. However, it's still very possible that not every change is noted, and even if a change is noted, it could only be included with some lag. Additionally, although I can use variation in road upgrading, this can only be observed when a road changes categories. That is, road maintenance or changes that would not count as upgrades across categories (such as pothole filling) are not captured. This may be a particular issue in the more recent years, as it is expected that a greater proportion of road spending reflects this unobserved variation.

3 A model of place effects in spatial equilibrium

The first empirical challenge associated with understanding how changes in road building influence local opportunity is that of measurement. How should one measure the effect a given road would have on local connectivity, both in the locations the road connects and in those it does not? The influence of any given road will depend on its position within the entire road network and the pre-existing distribution of economic activity.[5]

To overcome this complexity, I embed the simple place effects framework of Chetty and Hendren (2018a) into a quantitative spatial economics model of education completion, building on Hsiao (2024) and Allen et al. (2024). I take as my point of departure the place-effects literature, where outcomes, Y_k, for individuals k depends on factors specific to k and those specific to where k grew up, denote by $i(k, a)$, the location where k was at age a. In this paper, I am interested in the outcome Y_k, a dummy variable that indicates whether or not an individual has completed primary school. Following Chetty and Hendren (2018a) and others, I assume that place and individual effects are additively separable. Given this, I can decompose outcomes into that due to place effects and that due to individual effects as in equation 1.

$$Y_k = \sum_a \mu_{i(k,a)} + \theta_k \tag{1}$$

Place effects are denoted by μ_i where i is a given location, and θ_k captures place-invariant individual-specific effects. In this section, I provide a structural interpretation for this expression in the case where the outcome of interest is schooling. To do this I develop a quantitative spatial economics model of education and location choice. This model follows Hsiao (2024), extending his framework by endogenising education demand in spatial equi-

[5] We can gain intuition on this potential complexity by considering a simplified case. In a three-location economy, the impact on location A of a road connecting locations A and B will depend on the size of the potential market in B. It will also depend on any induced A to B, or B to A, migration. Finally, location C plays a role; perhaps trade is diverted away from C, causing an exodus of workers to A and B. Alternatively, the road could improve outcomes in B, which in turn increases trade from C, causing a relative decline in A.

librium.

Timing and dynamics.

Individuals are born in a given period t in an origin location i. They will stay and be educated in i before choosing a location j to move to in adulthood.[6] Individuals receive education shocks ξ_k and a vector of location-specific skill shocks $\{\varepsilon_{jk}\}$. They choose education e_k after observing their education shock, but before observing their skill shocks, they then observe skill shocks before choosing their migration location. After migrating each migrant has one child who will be educated in their parents migration location.

This framework displays cross-generation dynamics because the pervious periods migration patterns determine the next periods young who become educated and themselves choose where to migrate. However, within an individual, all choices happen within the same period, avoiding the need to consider dynamic choices. I will also make the simplifying assumption that individuals do not consider the utility of their unborn child when considering education choices or migration decisions. The dynamics in this framework are therefore those of spatial equilibrium and will be captured entirely by population movements.

Education.

Individuals choose e_k to maximise expected future utility during work, subject to education costs.

$$e_k^* = \arg\max_e \{\bar{v}_{it}(e) - c_{it}(e, \xi_k)\} \tag{2}$$

Where $\bar{v}_{it}(e)$ is the future expected utility, and $c_{it}(e, \xi_k)$ is the cost of education in location i in period t and is subject to the education shock ξ_k. I parameterise costs as a linear function of education $c_{it}(e, \xi_k) = e \cdot \tau_{it}^{educ} \cdot \xi_k$. Individuals move in adulthood to the location that maximises their utility subject to location-specific shocks ε_{jk}, therefore: $\bar{v}_{it}(e) = \mathbb{E}[\max_j v_{ijt}(e, \varepsilon_{jk})|e]$.

Utility.

Individuals with education e gain utility from working and living in a given location j in period t, which comprises of three components. First, individuals gain utility from local amenities a_{jt}, second from real wage income w_{jt}/P_{jt}, and thirdly are subject to iceberg migration costs $(\tau_{ijt}^m)^{-1}$. Real wage income is given by $w_{jt}/P_{jt} = \frac{r_{jt}}{P_{jt}} h_{jt}$ where r_{jt}/P_{jt} is the real wage per unit of human capital and h_{jt} is the quantity of human capital acquired. This is a non-linear function of education completed $h_{jt} = e^{\eta}\varepsilon_j$, where ε_j are the location-specific skills shocks that will be resolved before choosing location and are distributed as a type two

[6]Therefore, in this simple version of the theory, I restrict equation 1 to the single-location setting.

extreme value distribution with parameter θ. The utility of an individual born in i, moving to j can therefore be written as follows.

$$v_{ijt} = a_{jt} \cdot \left(\frac{r_{jt}}{P_{jt}} e^{\eta} \varepsilon_j \right) \cdot \left(\tau^m_{ijt} \right)^{-1} \tag{3}$$

Using this equation and leveraging properties of type two extreme value distributions I can therefore write an expression for expected utility.

$$\bar{v}_{it}(e) = \mathbb{E}[\max_j v_{ijt}(e, \varepsilon_j)|e] = e^{\eta} \sum_j \pi_{ijt} \frac{a_{jt} r_{jt}}{P_{jt} \tau^m_{ijt}} \mathbb{E}[\varepsilon_j | j] \tag{4}$$

Where π_{ijt} is the probability that an individual moves to j from i in period t. By considering the migration choice problem this object can be characterised.

Migration.

Individuals choose location j to migrate to, taking education e as given. As ε_j has a type two extreme value, the utility representation given in 3 results in the familiar gravity formulation for migration probability π_{ijt}.

$$\pi_{ijt} = \frac{\left(a_{jt} \frac{r_{jt}}{P_{jt}} (\tau^m_{ijt})^{-1} \right)^{\theta}}{\sum_j \left(a_{jt} \frac{r_{jt}}{P_{jt}} (\tau^m_{ijt})^{-1} \right)^{\theta}} \tag{5}$$

Note that using properties of the type two extreme value distribution, I can also write $\mathbb{E}[\varepsilon_j|j] = \gamma \pi_{ijt}^{-1/\theta}$, where $\gamma = \Gamma(1 - \frac{1}{\theta})$ and $\Gamma(\cdot)$ is the Gamma function. Putting this together with equation 5, equation 4 can be simplified to the following.

$$\bar{v}_{it}(e) = \gamma e^{\eta} \left(\sum_j \left(a_{jt} \frac{r_{jt}}{P_{jt}} (\tau^m_{ijt})^{-1} \right)^{\theta} \right)^{\frac{1}{\theta}} = \gamma e^{\eta} MA_{it}^{\frac{1}{\theta}} \tag{6}$$

Where market access is defined as the local availability of high utility (high amenity and or real wage) locations $\text{MA}_{it} = \sum_j \left(a_{jt} \frac{r_{jt}}{P_{jt}} (\tau^m_{ijt})^{-1} \right)^{\theta}$. The benefits of education are therefore tied to the local availability of high returns to education. Local changes to connectivity will make it easier for individuals to move to high-return locations, and this is well captured within this framework. However, changing connectivity may also alter the distributions of local returns itself. To capture this force, of opportunity moving, I endogenise local wages and prices.

Endogenous local real wages.

Firms produce location-specific differentiated goods under perfect competition but are subject to iceberg trade costs denoted by τ^t_{ijt}. Individuals have CES preferences over local varieties with an elasticity of substitution σ. Human capital is the only factor of production. Perfect competition implies that price equals marginal cost, $p_{jt} = r_{jt}/B_{jt}$ where B_{jt} is j, t-specific productivity. CES preferences and iceberg trade costs imply a familiar gravity equation in trade, which in turn gives the usual price aggregator $P_j^{1-\sigma} = \sum_i (\tau^t_{ijt} r_{it}/B_{it})^{1-\sigma} = TMA_{jt}$. Similarly, assuming symmetric trade costs, given human capital, and assuming goods markets clear, I have that $Y_{jt} = r_{jt} H_{jt} = p_{jt}^{1-\sigma} TMA_{jt}$. Solving the implied system I find $r_{jt} = (H_{jt}^{-1} TMA_{jt})^{1/\sigma} B_{jt}^{(\sigma-1)/\sigma}$. Where H_j is the total local stock of human capital in a given location.

Solving education choice.

Using the above derivations, the choice problem of an individual, k, can be solved to find the following.

$$\ln(e_k) = \alpha + \frac{1}{\theta(1-\eta)} \ln(\mathrm{MA}_{it}) - \frac{1}{1-\eta} \ln(\tau^{educ}_{it}) - \frac{1}{1-\eta} \ln(\xi_k) \tag{7}$$

Where $\alpha = \frac{1}{1-\eta} \ln(\gamma\eta)$. This equation encapsulates the place effects model. It can be rewritten in the form given in equation 1 by noting that place-effects are captured by the first two terms, $\mu_{it} = \frac{1}{\theta(1-\eta)} \ln(\mathrm{MA}_{it}) - \frac{1}{1-\eta} \ln(\tau^{educ}_{it})$ and individual-specific effects are given by the final term, $\theta_k = -\frac{1}{1-\eta} \ln(\xi_k)$. Those with higher education skill shocks (lower ξ_k) will complete more education irrespective of local conditions. Those who grow up in locations with higher market access or lower costs of education will complete more education, irrespective of their skill shock.

The sensitivity of educational outcomes to changes in market access depends on two parameters θ and η. The parameter $\theta > 0$ captures the dispersion of location-specific skill shocks. Larger θ implies less dispersed location-specific skill, and so higher migration sensitivity to changes in local (real) wages. Therefore, a higher θ implies a lower sensitivity of education choices to market access as the impact of any location-specific changes is smoothed out to a greater degree through increased migration. The parameter $\eta \in (0, 1)$ captures the concavity of the human capital production function. An η closer to 1 implies slower decreasing returns to education and thus greater sensitivity of education completion to incentives to educate, including changes to market access. With only variation in market access, these two channels cannot be separately identified, and instead only the bundle $\beta = (\theta(1-\eta))^{-1}$ can be identified.

4 Reduced form: The impact of market access on local opportunity

There are two key challenges associated with taking equation 7 to the data. First, measurement, I need to be able to measure a location's market access and how this changes over time. Second, identification. As τ_{it}^{educ} and ξ_k are not observed, but likely correlated with market access, a pair of strategies needs to be developed to allow β to be identified. Indeed, I can write equation 7 in a regression form as:

$$Y_k = \alpha + \beta \cdot \ln\left(\mathrm{MA}_{it}\right) + v_k \tag{8}$$

Where the error term is given by $v_k = -\frac{1}{1-\eta}\ln(\tau_{it}^{educ}) - \frac{1}{1-\eta}\ln(\xi_k)$. This formalisation allows a clear elucidation of the two key empirical challenges. First, individuals may select into better locations, causing a correlation between $\ln(\mathrm{MA}_{it})$ and $\ln(\xi_k)$. Second, locations with higher market access may also be conducive to education for endogenous reasons, causing a correlation between $\ln(\mathrm{MA}_{it})$ and $\ln(\tau_{it}^{educ})$, where note τ_{it}^{educ} can be interpreted as all unobserved factors relating to the cost (or benefit) of education in i in period t.

4.1 Measuring market access

The market access of a given location increases in the distance weighted power-sum of opportunities available in said location $\mathrm{MA}_{it} = \sum_j \left(a_{jt}\frac{r_{jt}}{P_{jt}}(\tau_{ijt}^m)^{-1}\right)^{\theta}$. However none of a_{jt}, r_{jt}, P_{jt} or τ_{ijt} are observed in the data. To overcome this, a location's market access can be written in a more convenient way. Note that $L_{jt} = \sum_i \pi_{ijt}L_{it} = \left(a_{jt}\frac{r_{jt}}{P_{jt}}\right)^{\theta}\sum_i(\tau_{ijt}^m)^{-\theta}L_{it}\mathrm{MA}_{it}^{-1}$. Denote $\widetilde{\mathrm{MA}}_{it} = \sum_j(\tau_{ijt}^m)^{-\theta}L_j\mathrm{MA}_{jt}^{-1}$. As shown by Donaldson and Hornbeck (2016) the only solution for this series of equations is that $\mathrm{MA}_{it} = a\widetilde{\mathrm{MA}}_{it}$ for some constant a. Thus, up to scale, $\mathrm{MA}_{it} = \sum_j(\tau_{ijt}^m)^{-\theta}L_{jt}\mathrm{MA}_{jt}^{-1}$. This expression can be taken to the data as L_{jt} is observed. To calculate market access only $T_{ijt} = (\tau_{ijt}^m)^{-\theta}$ needs to be estimated.

4.1.1 Calculating time-varying location-to-location travel times

I parameterise the iceberg travel costs as depending on i to j travel times over the road network prevailing at time t, denoted as t_{ijt}. These times are estimated using the prevailing road network at time t using Dijkstra's algorithm. Data on the road network is generated from historical road maps that I digitise, travel times over the four road categories are taken from Jedwab and Storeygard (2021). The online appendix section 3 details how t_{ijt} is calculated. I then parameterise these costs in a log-linear fashion such that $\ln(T_{ijt}) = -\theta \cdot \ln(\tau_{ijt}^m) = -\tilde{\theta} \cdot \ln(t_{ijt})$, where $\tilde{\theta} = b \cdot \theta$. Therefore, to calculate market access, it only remains to estimate $\tilde{\theta}$.

4.1.2 Estimating $\tilde{\theta}$

We can estimate $\tilde{\theta}$ using a standard gravity migration equation coupled with data from the censuses on locality-to-locality moves in each period. From the theory the number of individuals moving from i to j in period t can be written as $M_{ijt} = \pi_{ijt}L_{it}$ given the previously derived expression for π_{ijt} and including fixed effects results in the following framework given in equation 9. Where α_{it} and τ_{jt} denote origin-time and destination-time fixed effects respectively.

$$\ln(M_{ijt}) = -\tilde{\theta} \cdot \ln(t_{ijt}) + \alpha_{it} + \tau_{jt} + \varepsilon_{ijt} \tag{9}$$

This theory-consistent gravity equation can be estimated using pseudo Poisson maximum likelihood (Silva and Tenreyro, 2006; Yotov et al., 2016) to find $\hat{\tilde{\theta}} = \hat{\theta} \cdot \hat{b}$. By including this rich set of fixed effects this specification only uses variation in travel times stemming from bilateral travel costs. For example, a common threat to identification would be that as a location becomes more attractive, planners are more likely to build better connections to this location. Oppositely, planners might build connections to a previously flagging location in order to galvanise it. In either case, such behaviour would be captured by the fixed effects in this specification.

Any remaining threats to identification must operate at the bilateral level. One first-order concern of this type is that bilateral cultural ties between two locations may both influence migration and the strength of the road connection between these places. In this case cultural ties, denoted by ct_{ijt}, would be an omitted variable potentially biasing $\hat{\tilde{\theta}}$. Using data from Weidmann et al. (2010), I can control directly for the group similarity between locations.

More generally, one may be concerned that a planner builds better connections between two locations in order to facilitate existing or encourage new migration between these two places. If this were the case changes in travel time due to road building and upgrading maybe for reasons endogenous to migration rates. Moreover, preexisting strong connections maybe due to great social or economic ties, which may also encourage greater bilateral migration. To overcome these concerns I first consider estimating the gravity equation in the cross-section only, and second, consider using straight-line "as the crow flies" bilateral centroid distances as opposed to the estimated travel costs, which use the actual road network.

Table 1 shows the results from estimating equation 9 using the approaches described above. These estimates show remarkable stability across specifications. Due to this consistency the first column is taken as the preferred specification, leading to an estimate of $-\tilde{\theta} = -1.516$. Few estimates of this parameter are available in the literature from a developing country. Perhaps most informative Morten and Oliveira (2024) estimates an elasticity of -1.698 using meso-to-meso migration over a five year period in Brazil. The estimating sample and context are very different between these two settings, but it is nevertheless reassuring that both estimates are similar.

Table 1 Estimating the Migration Elasticity

	(1) PPML	(2) PPML	(3) OLS	(4) OLS	(5) PPML	(6) PPML	(7) PPML
Log(Travel time)	-1.516***	-1.494***	-1.418***	-1.417***	-1.482***	-1.510***	
	(0.0379)	(0.0388)	(0.0203)	(0.0203)	(0.0388)	(0.0460)	
Group Similarity					0.106	0.124	0.0140
					(0.110)	(0.119)	(0.128)
Log(Dist. Crow)							-1.244***
							(0.0411)
Destination-time FE	X	X	X	X	X	X	X
Origin-time FE	X	X	X	X	X	X	X
Sample	Panel	Panel	Panel	Panel	Panel	X-section	X-section
Winsorised		1 & 99		1 & 99			
N	26234	26234	13616	13616	26234	9450	9450

Notes: This table shows the results from estimating equation 9 using various specifications. All specifications use destination-time and origin-time fixed effects with standard errors clustered at the origin-destination pair level. Column one uses PPML, column two also uses PPML winsorising migration flows at the 1st and 99th percentile. Columns three and four instead use a log-linear OLS specification. Column five again uses PPML controlling for group similarity. Columns six and seven move to the cross-section, taking the first available year of data; in both of these columns, group similarity is controlled for. In column seven, instead of using travel times estimated from the actual road network, I use as-the-crow-flies centroid-to-centroid distances.

With estimates of $\tilde{\theta}$, in hand using bilateral travel times I can then calculate each locations market access: $\text{MA}_{it} = \sum_j t_{ijt}^{-1.516} L_{jt} \text{MA}_{jt}^{-1}$. This is a series of non-linear equations across locations, which can be efficiently solved using an iterative procedure and has a unique up-to-scale solution (Donaldson and Hornbeck, 2016). Table 1 in the online appendix shows that estimates of θ are stable across countries and years.

4.2 Identifying the reduced form effect of market access on local opportunity

To identify the causal effect of market access on local opportunity, I combine a movers design with a market access instrumental variables strategy that leverages plausibly exogenous sources of variation.

Movers Design.

The movers' design uses variation in exposure to different levels of market access by comparing individuals who moved to high market access locations earlier in their childhood to those who moved later and so were less exposed. This allows selection into high-quality (high market access) locations, but requires that selection does not vary systematically with the age at which the child moves. This is the standard assumption in this literature, I perform various robustness and validation exercises probing this assumption in section 4.4.

To operationalise the movers' design, I restrict my sample to one-time movers who are between the ages of 14 and 18 at the time of being surveyed, but could have moved at any age prior. In my sample, 81% of individuals in this age range have not moved, 13% have moved once, and 6% have moved more than once. I then make the simplifying assumption of a linear dose effect. Under this assumption what matters for an individual is their exposure to market access over their childhood. I compare children in the same cohort who move from the same origin location to destination locations with different levels of market access at different ages. Leveraging only variation in the age-at-move I can then estimate the causal effect of spending an additional year of childhood in a one percent higher market access location on the probability of completing primary school.

We can write equation 8 in a sample of one time movers between 14 and 18 as $Y_k = \frac{1}{18}\left(m_k * \mu_{o(k),c(k)} + (18 - m_k) * \mu_{d(k),c(k)}\right) + \theta_k$. Where $o(k)$ is individual k's origin location, $d(k)$ is their destination location, m_k is the age at move, and $c(k)$ is k's cohort. From now on, I will suppress implicit dependence on k. I can include origin by cohort fixed effects and write place effects in terms of differences to find $Y_k = \left(1 - \frac{m_k}{18}\right) \cdot \Delta\mu_{odc} + \alpha_{oc} + \theta_k$. Following the theory discussion, I then write the location effect for a given location l as $\mu_{lc} = \beta \cdot \mathrm{MA}_{lc} + v_{lc}$, where v_{lc} captures unobserved aspects of place effects. Substituting this into the above, I find $Y_k = \left(1 - \frac{m_k}{18}\right) \Delta\mathrm{MA}_{odc} + \alpha_{oc} + \tau_m + \varepsilon_k$, where $\varepsilon_k = \left(1 - \frac{m_k}{18}\right) \Delta v_{odc} + \theta_k$ is unobserved. The conditioning on $\Delta\mathrm{MA}_{odc}$ in levels, including age-at-move fixed effects, and finally allowing effects to differ before vs after moving at age 13 gives me the specification in 10. Following Milsom (2023), and Chetty and Hendren (2018a), I allow differing intercept and slope coefficients for those who move before age 14 and those who move after (inclusive). Those who move after 14 have already completed primary school, and so market access should not affect their outcomes. The coefficient of interest is β_1, which can be interpreted as the impact of spending an additional year of childhood in a 100% higher market access location on the probability of completing primary school.

$$\begin{aligned} Y_k = \mathbb{1}_{[m_k \leq 13]} \cdot (\alpha_1 + \beta_1 \cdot (18 - m_k)) \cdot \Delta\mathrm{MA}_{odc} + \\ \mathbb{1}_{[m_k > 13]} \cdot (\alpha_2 + \beta_2 \cdot (18 - m_k)) \cdot \Delta\mathrm{MA}_{odc} + \\ \tau_m + \alpha_{oc} + \varepsilon_k \end{aligned} \tag{10}$$

By specifying the error term ($\varepsilon_k = \left(1 - \frac{m_k}{18}\right) \Delta v_{odc} + \theta_k$), the two potential sources of bias can be seen clearly. First, correlation between $m_k \Delta \text{MA}_{odc}$ and θ_k will cause bias. This occurs exactly when systematically "better" or "worse" individuals sort into higher or lower market access locations systematically earlier or later. That is, as with normal mover designs, sorting into better locations is permissible, but not that this happens systematically differently at different ages-at-moves. The assumption maintained here is weaker than the usual, as here I only require no such selection into locations on one characteristic — market access. The second potential source of bias is correlation between ΔMA_{odc} and Δv_{odc}. This will occur if higher market access locations are better on some other unobserved dimension, i.e. if there is a correlation between MA_{lc} and v_{lc}. This seems possible in the case of road building, as roads are not exogenously placed. The planner could build roads to service booming locations or to galvanise flagging ones, and market access terms will inherit this endogeneity.

Not-on-least-cost-path variation.

To overcome the endogeneity of road placement and therefore market access, I require an additional identification strategy and leverage the construction of market access terms to consider only not-on-least-cost-path variation.

Previous identification strategies designed to overcome the endogeneity of road placement include using placebo lines from planned but unbuilt routes Donaldson (2018); Okoye, Pongou, and Yokossi (2019), using straight line or least-cost path spanning tree instruments[7] Moneke (2020); Michaels (2008); Ghani, Goswami, and Kerr (2016); Faber (2014), or leveraging *far-away* variation in road changes Donaldson and Hornbeck (2016); Jedwab and Storeygard (2021). In my setting, it is difficult to see how the first two approaches can be implemented. First, I don't have data on unbuilt but planned placebo lines. Second, there is no clear set of locations that are being connected, and in addition, a significant proportion of the variation in travel times comes from road upgrading rather than the building of entirely new roads, in this setting, it's unclear how localities are "incidentally" connected.

The third strategy, leveraging far-away variation in roads, is appealing but suffers from a number of known drawbacks. First, it's unclear how far "far-away" should be; and although researchers can present many distances, it is ultimately an ad-hoc choice. Second, and more fundamentally, variation due to large projects which may be far away but for endogenous reasons, or relatively far away connections that are built to ease transport to, or encourage trade to a given location, remain threats to identification.

In this paper, I propose a novel identification strategy that builds upon the far-away

[7]Locations that just happen to lie between two cities that are being connected by a road may be plausibly described as exogenous. This is also known as the *inconsequential units* IV.

variation approach by considering not-on-least-cost-path variation. The not-on-least-cost-path variation approach only uses changes in a locality i's market access that stems from indirect changes to all other locations' market access freezing migration costs along the least cost path from i to all other locations.[8] This approach has two intuitive explanations. First, one can consider it as the same as using far-away variation, but whereas far-away variation defines distance over Euclidean space, not-on-least-cost-path defines distance over network space, where a "distance" of one refers to one-degree removed indirect variation. Under this interpretation, the network structure is taken seriously and resolves the ad-hoc nature of what "far-away" may mean by appealing to the theory. Secondly, one can think of not-on-least-cost-path variation as approximating the decision-making process of the policymaker building roads, and using the residual variation. If a central planner builds roads to or from i in order to directly improve its connectivity, this variation is not used, and instead the residual variation in market access is leveraged.

To formalise the above discussion, consider a generic market access variable $\text{MA}_{it} = \sum_j (\tau_{ijt}^m)^{-\theta} L_{jt} (\text{MA}_{jt})^{-1}$. Following Donaldson (2018), I first remove own-location market access (therefore sum over other locations $j \neq i$) and freeze the market size variable at the initial level (L_{j0}) as these objects are co-determined with the outcome variable. Then I can use the not-on-least-cost-path variation (freeze the least cost path to the initial value τ_{ij0}^m) of degree n by constructing the following instrument.

$$\text{MA}_{it}^{IV} = \sum_{j \neq i} (\tau_{ij0}^m)^{-\theta} L_{j0} \left(\text{MA}_{jt}\right)^{-1}$$

Where $\text{MA}_{it} = \sum_{j \neq i} (\tau_{ijt}^m)^{-\theta} L_{j0} \left(\text{MA}_{jt}\right)^{-1}$ gives actual market access terms.

The instrument can be re-written as a shift-share instrument where the shares are given by $s_{ij} = (\tau_{ij0}^m)^{-\theta} L_{j0}$ and the shifts are given by $g_{jt} = \text{MA}_{jt}^{-1}$, then I can write $\text{MA}_{it}^{IV} = \sum_{j \neq i} s_{ij} \cdot g_{jt}$.[9] The identifying assumption is that, conditional on fixed effects and controls (including region×year), these destination network shocks are orthogonal to unobserved

[8] This approach is not to be confused with one variation of the incidental connection approach which uses an estimated least cost route over an estimated cost surface to instrument for actual connections. I don't take this approach in this paper for the same reasons that incidental connection approaches, in general, are unsuitable, and in addition, because a significant proportion of the variation comes from within road quality, and it is unclear how to leverage this intensive margin dimension using this approach.

[9] This formulation allows us to clarify where the variation comes from by decomposing changes in market access into the components given below.

$$d \ln(\text{MA}_i) = \sum_{j \neq i} s_{ij} \cdot d \ln \left((\tau_{ijt}^m)^{-\theta} \right) - \sum_{j \neq i} s_{ij} \cdot d \ln(\text{MA}_j)$$

The first term of the right-hand side gives the direct effect of changes to the network on market access in i, whereas the second term gives the indirect network-mediated effect. It is this second term that is captured by the not-on-least-cost-path instrument, which effectively purges market access of the direct effect.

determinants of outcomes in i, except through their impact on i's market access. This helps clarify the main threats to identification. First, baseline exposure may be endogenous. To overcome this, I follow the approach of Borusyak and Hull (2023) and re-centre the instruments; details of this procedure are in sub-section 4.4. Second, non-modelled spillovers. Other shocks in j could affect i proportionally to s_{ij}, such as political economy shifts, or other public infrastructure correlated with roads. Similarly, regional policy combining road building and other projects would cause correlation across local shocks and i's outcomes. These threats are dealt with by (1) combining the not-on-least-cost-path approach with the far-away variation approach to remove variation local to i, and (2) including region by year fixed effects in our analysis. I also directly control for related concerns surrounding clientelism.

To validate the reasonableness of the arguments made in this section I provide some Monte-Carlo simulation evidence presented in table 2. I create a simulated network of over 1,000 nodes. Each node lies on the unit square and belongs to one square quadrant, denoted as a region. I randomly generate an undirected graph over these nodes, where each location is directly connected to its six nearest neighbours, and there is a small probability of a random long-link connection between any two randomly chosen nodes. Over this "road" network, I then calculate the travel time between each pair of nodes along connected links as $\omega_{ij} = ||s_i - s_j|| + 0.05$ where s_i are the two-dimensional coordinates of location i. This is converted into a travel cost using $T_{ij} = d_{ij}^{-2}$, where d_{ij} is the shortest path from i to j along the network. This creates the baseline network. Each node is also given a random baseline population $L_i \sim LogNormal(0, 0.5^2)$ and the sum of population is normalised to one.

To this I construct a road-building shock on a random subset of edges according to two data-generating processes. The first incorporates endogenous road upgrading. Let location i have some unobserved characteristic $u_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Then each baseline edge (a, b) is upgraded with probability increasing in u_a and u_b and equal to $\mathbb{P}(\text{upgrade } (a, b)) = 0.18\frac{p_a+p_b}{2}$ where $p_i = \Lambda\left(1.8\frac{u_i-\bar{u}}{\sigma_u}\right)$ and $\Lambda(x) = \frac{1}{1+e^{-x}}$. If a road segment is upgraded, its travel time is multiplied by 0.6. The shortest travel times between each pair of nodes are then calculated over this new, randomly upgraded, network. The second data-generating process follows the first, but additionally introduces some regional building program where for some region r, if the start and end point of the road both fall within r, there is an additional probability that the road is upgraded given by 0.33.

In each location under each scenario I then calculate market access terms and market access using the not-on-least-cost-path instrument, using far-away variation (only variation beyond the 20th percentile of distances), and both. Let $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 0.25^2)$ be a random shock. I then construct two data-generating processes for the location outcome y_i following the two processes for road upgrades discussed above.

$$\text{DGP-A:} \qquad y_i = 1 \cdot \log(\text{MA}_i) + 2 \cdot u_i + \varepsilon_i$$

$$\text{DGP-B:} \qquad y_i = 1 \cdot \log(\text{MA}_i) + 2 \cdot u_i + 2 \cdot \mathbb{1}_{[r(i)=r]} + \varepsilon_i$$

The true coefficient on log market access is therefore one. Using these data-generating processes, I regress y_i on log market access using OLS, using OLS plus region fixed effects, using the far-away IV, using the not-on-least-cost-path IV, using the NLCP and far-away IV, and using the NLCP IV and region fixed effects. Table 2 shows the results averaged over 250 simulations.

Panel A of table 2 shows the performance of the various estimation strategies when the endogeneity comes from correlated local characteristics. The OLS coefficient is very biased (150%), and including regional fixed effects does not attenuate this bias. Focusing on far-away variation reduces the bias to 50%, but does not eliminate it. However, using a not-on-least-cost-path approach significantly reduces the bias to around 8%. Panel B then turns to the second data-generating process, which includes a regional development program. Here, naive OLS is hopelessly biased, although including regional fixed effects does reduce this bias it remains at 133%. The far-away variation approach is not helpful here, nor is only using the not-on-least-cost-path approach — both have bias similar to naive OLS. Including both not-on-least-cost-path and far-away variation deals with both sources of endogeneity fairly well and exhibits a bias of 33%. However, the most successful approach is simply to condition on region fixed effects when using the not-on-least-cost-path instrument, which has a bias of 9%. In sum, the simulation results show that the not-on-least-cost-path approach is the most successful in dealing with the non-random placement of roads. However, it is not without caveats, and conditioning on regional fixed effects is important when dealing with regionally correlated development programs.

Table 2 Not-on-least-cost path simulation evidence

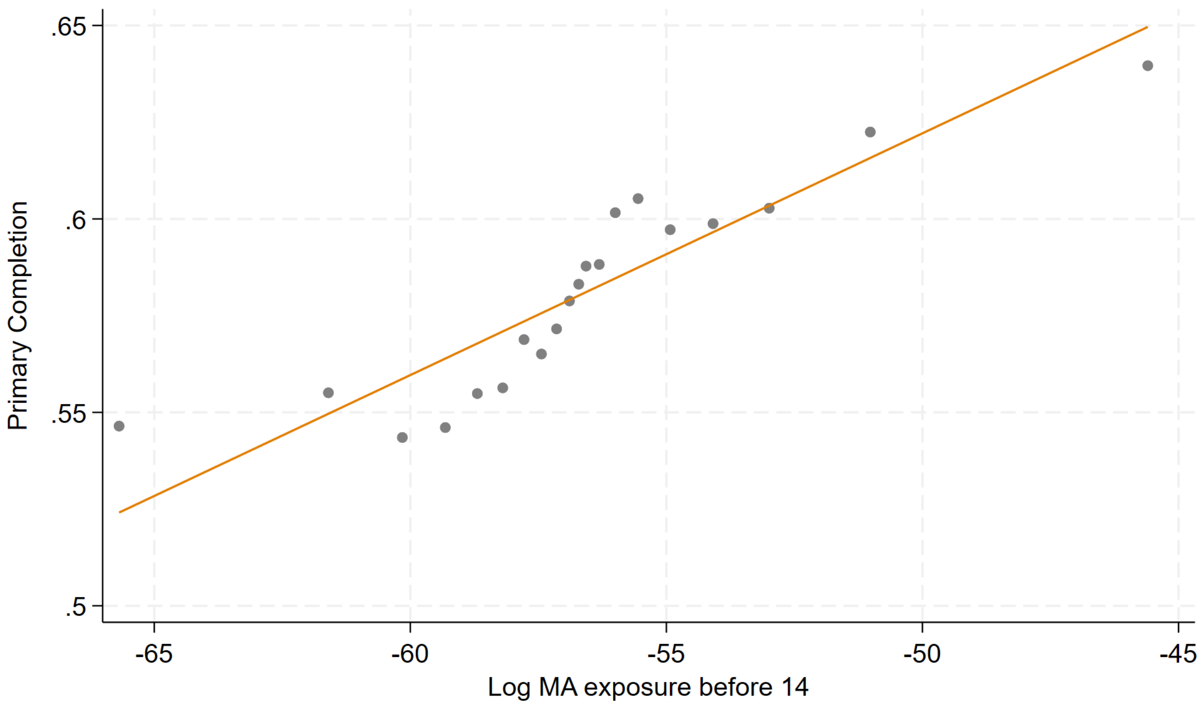
Estimator	Mean	SD	Bias
DGP A: Local Targeting			
OLS	2.554	0.428	1.554
OLS + Region FE	2.564	0.435	1.564
2SLS: Far-away IV	1.495	0.863	0.495
2SLS: NLCP IV	0.919	0.506	-0.081
2SLS: NLCP + Far-away variation IV	0.933	0.930	-0.067
2SLS: NLCP IV + Region FE	0.921	0.517	-0.079
DGP B: Local and Regional Targeting			
OLS	4.319	0.366	3.319
OLS + Region FE	2.328	0.396	1.328
2SLS: Far-away IV	4.101	0.859	3.101
2SLS: NLCP IV	-3.480	1.468	-4.480
2SLS: NLCP + Far-away variation IV	1.325	1.279	0.325
2SLS: NLCP IV + Region FE	0.909	0.505	-0.091

Notes: This table reports Monte Carlo evidence for the market-access instruments described in the text. Each simulation draws a network of 1,000 nodes on the unit square (nodes assigned to one of four regions), baseline populations, and road-upgrading shocks; outcomes are generated under two data-generating processes: (A) $y_i = \log(\text{MA}_i) + 2u_i + \varepsilon_i$ (endogenous upgrading through u_i) and (B) $y_i = \log(\text{MA}_i) + 2u_i + 2\mathbb{1}[r(i) = r] + \varepsilon_i$ (endogenous upgrading plus a region-level program). For each draw, I estimate the coefficient on $\log(\text{MA}_i)$ using: OLS; OLS with region fixed effects; IV using "far-away" variation (constructing exposure using only destinations beyond the 20th percentile of network distances); IV using the not-on-least-cost-path (NLCP) instrument, i.e., holding bilateral trade costs at baseline least-cost paths (τ^m_{ij0}) and using shocks $g_{jt} = \text{MA}^{-1}_{jt}$; IV combining NLCP and far-away restrictions; and NLCP-IV with region fixed effects. Entries are averages across 250 simulations. SD refers to the standard deviation across simulations.

4.3 Results

Figure 2 shows the binscatter relationship between primary school completion rates and log market access exposure before age 14, controlling for origin by cohort and age at move fixed effects, but without following a movers design or tackling the endogeneity of market access. Nevertheless, this figure shows a strong positive and approximately linear association, individuals who were more exposed to higher market access during the first 13 years of their childhood were more likely to complete primary education.

Figure 2 Binscatter relationship without instrumenting



Notes: This figure shows the non-parametric relationship between market access exposure before 14 and primary completion in a bin-scatter figure, conditional on origin by cohort and age at move fixed effects.

Figure 3 shows the results from estimating equation 10 on a sample of one-time movers between the ages of 14 and 18. It displays coefficients $\widehat{\beta}_1$ and $\widehat{\beta}_2$ as well as their 95% confidence intervals. Each colour and marker shape corresponds to a different empirical strategy. Each specification includes origin location by year born by census year by age fixed effects and age-moved fixed effects, and standard errors are clustered at the origin-year born level. In blue circles I plot the baseline results from regressing primary school completion on log market access. In red diamonds, I use the same specification, but instrument market access with the not-on-least-cost-path instrument. In green triangles, I then employ the mover's design described in equation 10. In yellow squares and purple hollow circles, I repeat this specification, instrumenting the change in market access with the not-on-least-cost path strategy and the not-on-least-cost-path plus far-away variation strategy.

Figure 3 Reduced form results

Notes: This figure shows the results from the reduced form analysis. All regressions are run on a sample of 14 to 18-year-olds and include origin location by year-born by census-year by age fixed effects, and age moved fixed effects. Standard errors are clustered at the origin by year of birth level, and indicated by horizontal lines. In blue with a circular marker, I show the baseline results of regressing average log market access over one's childhood on primary completion. In red with a diamond marker, I instrument market access using not-on-least-cost-path variation as discussed in the text (the first stage Kleibergen-Paap rank LM statistic is 458.6). In green with triangular markers, I then report slope coefficients from estimating the movers design given in equation 10. These coefficients refer to movers before and after age fourteen as indicated on the x-axis. In yellow with square markers, I then combine the instrumental variables approach with the movers design (the first stage Kleibergen-Paap rank LM statistic is 243.7). Finally, in purple, I use a stricter instrument that combines the not-on-least-cost-path approach with that using far-away variation with a 50km cut off (the first stage Kleibergen-Paap rank LM statistic is 296.1).

Across all specifications, Figure 3 shows a clear positive impact of exposure to high market access on primary completion rates before 14, and a reassuringly null impact after. Coefficients can be interpreted as the percentage point impact of spending an additional year in a one percent higher market access location. Therefore, the overall impact for non-movers in a given location of a one percent increase in market access is a $0.005 \times 13 = 0.07$ percentage point increase in the probability of completing primary education. The standard deviation of log market access is about one, so moving to a one standard deviation higher market access location at birth increases the probability of completing primary school by 7 percentage points or 12% of the average primary completion rate, which is 58% in my estimating sample.

4.4 Robustness and validity checks

In this section, I discuss four robustness and validity checks of the above results. Details of these tests and results can be found in the online appendix section 7.

First, the movers' design relies on the assumption that selection effects do not vary with the age at move. This cannot be tested in general; however, I can test whether selection on some observable characteristics varies with the age at move. To do this, I compare the "quality of move" measured as the change in location market access for those whose mothers have, and do not have, primary education, at different ages. If individuals with educated mothers systematically moved earlier (or later) to higher market access locations relative to those with non-educated mothers, this would be in violation of the identifying assumption. Figure 11 in the online appendix section 7 plots these results and finds no difference between those with and those without educated mothers.

Second, I include household fixed effects in equation 10, therefore only focusing on variation in age-at-move within a household across siblings. This specification obviates concerns around time-invariant household characteristics driving the observed relationship. Figure 12 in online appendix 7 shows the results. When including household fixed effects, I find that $\widehat{\beta}_1$ is slightly attenuated, but is far from being significantly so in either an economic or statistical sense.

Third, I show the robustness of the main result to varying specifications. Figure 13 shows that similar results are found if (i) I don't include origin by year born fixed effects but only year born fixed effects, (ii) I replace origin by year born fixed effects with origin fixed effects only, (iii) I remove age at move fixed effects, and (iv) I perform a PPML regression in levels.

Finally, I control for the possible non-random exposure to exogenous shocks baked into a market access type design Borusyak and Hull (2023). Intuitively, this problem arises when, under random road placement, some locations will still expect to have higher or lower market access due to, for example, their initial location in the network (central locations would mechanically see higher gains in market access). To overcome this, I follow Borusyak and Hull (2023) and specify a data-generating process for market access. I construct 250 random road networks starting from the baseline in 1970 and randomly upgrading roads until the actual country-year change in travel time has been reached. I then calculate each location's market access over each random network in each period and use the average of these terms, a location's *expected* market access. I then control for this expected market access and repeat the analysis shown in Figure 3. The online appendix section 1 shows the results from this exercise in figure 1 — coefficients do not qualitatively or quantitatively change.

In addition to the above, one may have setting-specific empirical concerns. In the online appendix section 8, I show that in this setting, clientelism is not a threat to identification

Burgess et al. (2015). In the online appendix section 9, I show that top-coding due to primary completion rates nearing 100% is not a concern. Finally, in the online appendix section 10, I show that the prevalence of Koranic schools or Medersas remains too low to be of empirical relevance in my setting.

5 Quantification and mechanisms

Section 4 recovered the reduced form impact of changes in connectivity on local opportunity. However, to recover the aggregate impact of road building since 1970, I need to know the counterfactual market access under the scenario where no roads had been built. Denote by a prime counterfactual variables, then our object of interest is: $\text{MA}'_i = \sum_j \left(\tau_{ij}^{m'}\right)^{-\theta} L'_j(\text{MA}'_j)^{-1}$. To calculate this quantity, I need to know the counterfactual population distribution L'_j, which in turn depends on the full structure of the model.

To recover this object, I will proceed in four steps. First, I derive a parsimonious non-linear system of equations in each location that determines all endogenous variables. Second, to avoid backing out location fundamentals, I solve this system in differences using exact hat algebra. Third, I identify the remaining model parameters. Fourth, I check for the existence and uniqueness of equilibrium and solve the system for the given counterfactual road network.

5.1 Solving the system

We can solve this system using exact-hat algebra into one of three endogenous variables in three equations in each location where hats denote counterfactual changes. These three endogenous variables are (1) the stock of human capital in a given location $\hat{H}_{it}$, (2) market access $\widehat{\text{MA}}_{it}$ and (3) goods market access $\widehat{\text{TMA}}_{it}$. The resulting expressions are given below; details of the derivation can be found in the appendix section 11.

$$\widehat{H}_i^{1+\frac{\theta}{\sigma}} = \widehat{\text{TMA}}_i^{\delta} \cdot \sum_j \omega_{ij} \cdot \left(\widehat{\tau}_{ij}^{m}\right)^{-\theta} \cdot \widehat{\text{MA}}_j^{\lambda}$$

$$\widehat{\text{TMA}}_i = \sum_j \rho_{ijt} \cdot \left(\widehat{\tau}_{ij}^{t}\right)^{1-\sigma} \cdot \widehat{\text{TMA}}_j^{\frac{1-\sigma}{\sigma}} \cdot \widehat{H}_j^{\frac{\sigma-1}{\sigma}}$$

$$\widehat{\text{MA}}_i = \sum_j \pi_{ij} \cdot \left(\widehat{\tau}_{ij}^{m}\right)^{-\theta} \cdot \widehat{\text{TMA}}_j^{\delta} \cdot \widehat{H}_j^{-\frac{\theta}{\sigma}}$$

Where $\lambda = \frac{\eta-\theta(1-\eta)}{\theta(1-\eta)}$ and $\delta = \frac{\theta(2\sigma-1)}{\sigma(\sigma-1)}$. The quantities $\omega_{ij}, \rho_{ij}, \pi_{ij}$ denote the initial shares of human capital, trade, and migration which are observed. Given counterfactual changes in the road network and parameter estimates, this system can therefore be solved for the endogenous variables, including the counterfactual change in local opportunity $\hat{\mu}_i$. In the appendix section 11, I derive conditions for the existence and uniqueness of the equilibrium of this system, following Allen et al. (2024). The parameter estimates used in the counterfactual exercise satisfy this condition.

5.2 Identification of model parameters

We need to identify parameters η, σ, θ and trade and migration iceberg costs $T^t_{ijt} = (\tau^t_{ijt})^{\sigma-1}, T^m_{ijt} = (\tau^m_{ijt})^{\theta}$. In section 4 T^m_{ijt} was estimated, and a bundle of η and σ was identified. Therefore, it only remains to identify η, σ, and θ separately as well as T^t_{ijt}.

5.2.1 Estimating trade costs T^t_{ijt}

As with T^m_{ijt} I can parameterise trade costs such that $T^t_{ijt} = (\tau^m_{ijt})^{\sigma-1} = t^{\tilde{\sigma}-1}_{ijt}$ where $\tilde{\sigma} = a \cdot \sigma$. Although data on migration is remarkably rich, covering a large time span and granular geography, allowing estimation of T^m_{ijt} with confidence — no analogous data exists for trade across space in this setting. As a result of this, I am forced to calibrate σ from the literature. I follow Morten and Oliveira (2024), who leverage variation in trade across Brazilian states. Morten and Oliveira (2024) calculate the elasticity of trade and migration costs to travel time. I take their estimated elasticity of trade to travel time (1.96) and rescale it by the ratio of my estimated migration costs to theirs (0.58) to account for differences in geography and environment. Following this method I find $\tilde{\sigma} = 1 + 1.96 \times 0.58 = 2.14$. Calibrating σ from Simonovska and Waugh (2014) to be $\sigma = 5$, I back out $a = 0.43$ and an elasticity of trade costs to travel times: $(-1.96 \times 0.58)/(1 - 5) = 0.28$.

5.2.2 Separately identifying θ and η

Following the strategy employed by Bryan and Morten (2019), I use model-implied regressions to separately identify θ and η. Note that the expected wage individuals from i who move to j attain is given by $\mathbb{E}[w_{jt}|i] = \gamma r_{jt} h_{it} \pi^{-1/\theta}_{ijt}$ an expression that motivates the following regression.

$$\ln(\bar{w}_{ijt}) = \alpha_{it} + \gamma_{jt} - \frac{1}{\theta}\ln(\pi_{ijt}) + v_{ijt} \tag{11}$$

This regression will therefore allow me to find estimates for θ which, coupled with our previous estimates of $\frac{1}{\theta(1-\eta)}$, will allow me to estimate η. However, estimates from equation 11 may be biased. As discussed in Bryan et al. (2014), any shock that affects the wages that individuals from i receive in j will also impact migration from i to j, for example, if demand for skills generated in i increases in j. To overcome this I use the instrumental variable approach suggested by Bryan and Morten (2019) and instrument $\ln(\pi_{ijt})$ with $\ln(\pi_{\neg ijt})$, where $\pi_{\neg ijt}$ is the proportion of people from other origins who migrate to j in period t.

Table 3 shows the results from estimating equation 11 with various specifications. Table 3 also shows the implied values of $\widehat{\theta}$ and $\widehat{\eta}$, combining information with that from estimating equation 10 with corresponding standard errors calculated using the Delta method. Column two is our baseline specification where $\ln(\pi_{ijt})$ is instrumented as discussed above. I find the following parameter estimates $\theta = 22.64\ (2.214), \eta = 0.434\ (0.170), \sigma = 5$. This compares to Hsiao (2024) who finds $\theta = 20, \eta = 0.224$, and Bryan and Morten (2019) who find $\theta = 28$.

Table 3 Identifying θ and η

	(1)	(2)	(3)	(4)
$\ln(\pi)$	-0.0411***	-0.0442***	-0.0447***	-0.0473***
	(0.00356)	(0.00432)	(0.00425)	(0.00432)
θ	24.30	22.64	22.36	21.15
	(2.11)	(2.21)	(2.12)	(1.93)
η	0.47	0.43	0.43	0.39
	(0.16)	(0.17)	(0.17)	(0.18)
IV		X	X	X
F-stat		3247	3379	3349
FE	ot,dt	ot,dt	o,d,t	o,d
Obs	13999	13999	13999	13999

Notes: This table shows the results from estimating equation 11 and the implied values of $\widehat{\theta}$ and $\widehat{\eta}$ combining information with that from estimating equation 10 with corresponding standard errors calculated using the Delta method. Column (1) shows results with no instrumenting, including origin-time and destination-time fixed effects. Columns (2) to (4) show results instrumenting following Bryan and Morten (2019) with various fixed effects specifications. Column (2) includes origin-time and destination-time fixed effects and is our baseline specification, column (3) includes origin, destination, and time fixed effects and finally column (4) only includes origin and destination fixed effects.

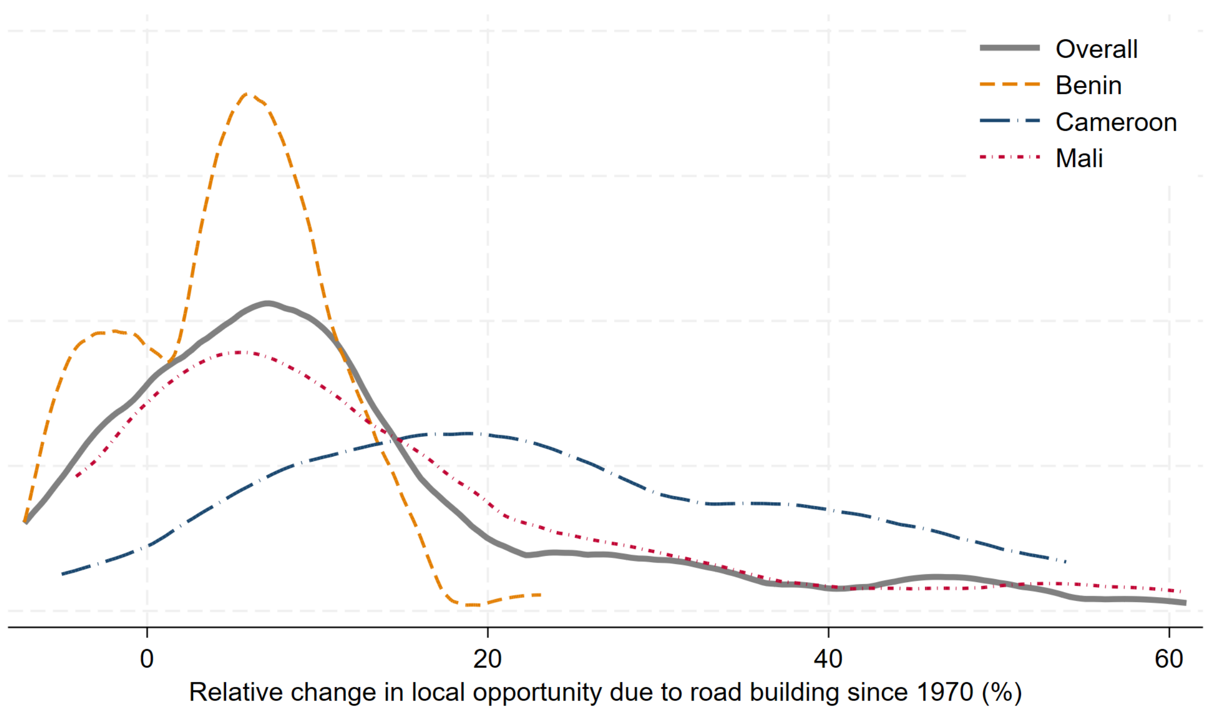
5.3 Quantification results

Using the full structure of the model, I can quantify the causal effect of road building between 1970 and 2020 on the distribution of local opportunity. I find that, on average, road building increased local opportunity. Due to road building since 1970, the causal effect of place on primary completion rates increased by, on average, 11.5%. This corresponds to a 0.22 percentage point higher average annualised growth rate. Over the same period across Benin, Cameroon, and Mali, the average annualised growth rate in primary education completion was 1.82 percent.[10]

[10] This average weights each location equally. Data from the World Bank available here.

The average hides considerable variation within-country across locations, with 17% of locations seeing aggregate negative effects. Figure 4 shows graphically how this average effect varies across locations. In each country, there is significant variation in the impact of road building since 1970 on local opportunity. In Cameroon, the distribution is relatively flat with a high average of 15.4%. Only three locations (8%) in Cameroon experience negative effects, whereas ten locations (26%) experience effects in excess of a 33% increase. Benin and Mali, on the other hand, have narrower distributions with means of 4.6% and 13.2%, respectively. In Benin 19 locations (25%) saw negative impacts, whereas in Mali, 6 locations (13%) did. In Benin the largest positive impact was an increase of 23%, whereas in Mali, the right tail extends to a 61% increase, and five locations (11%) experience an increase in opportunity of at least 33%.

Figure 4 Aggregate effects by country



Notes: This figure plots the empirical distribution over locations (weighted evenly) of the aggregate effect of road building since 1970 on local opportunity. In solid grey I plot the average over all countries, in dashed orange for Benin, in dot-long-dashed blue for Cameroon, and in dot-short-dashed red for Mali.

5.4 Mechanisms: Moving to Opportunity or Moving Opportunity

Changes to the road network alter the spatial distribution of local opportunity through two main channels. First, opportunity will increase because it is now easier for individuals to move to areas with pre-existing high opportunity. Second, the spatial distribution of opportunity itself may change. This latter force can be further decomposed into the endogenous reaction of the population and the endogenous reaction of goods trade. Endogenous migration will be negative, as more immigration to high-opportunity locations causes more competition for such opportunities. The endogenous reaction of demand, however, is likely to be positive as the same influx of individuals increases demand for goods and so the opportunities available. This second channel, that of opportunity itself moving, could therefore work to amplify or mitigate the direct effect — the overall impact on a given locality will depend on the entire road network and distribution of economic activity and its position within it. In this section, I decompose the aggregate effects shown in Figure 4 into the constituent channels described above. This allows us to quantify the relative importance of "moving to opportunity" vs "moving opportunity" on an aggregate and individual location level, as well as highlight the crucial importance of endogenising both migration and trade responses in the model.

Formally, the impact of changes in connectivity on local opportunity can be decomposed by appealing to equation 7 which states that the local proportional change in opportunity is equal to $\hat{e}_i = \widehat{\text{MA}}_i^{\frac{1}{\theta(1-\eta)}}$ for location i. Using our derived expression for market access, $\widehat{\text{MA}}_i = \sum_j \pi_{ij} \cdot \left(\hat{\tau}_{ij}^m\right)^{-\theta} \cdot \widehat{\text{TMA}}_j^{\delta} \cdot \widehat{H}_j^{-\frac{\theta}{\sigma}}$, this effect can therefore be split into four components as shown in the expression below.

$$
\begin{aligned}
\hat{e}_i^{\theta(1-\eta)} - 1 = &\underbrace{\sum_j \pi_{ij} \left((\hat{\tau}_{ij}^m)^{-\theta} - 1\right)}_{\text{Direct}} \\
&+ \underbrace{\sum_j \pi_{ij} \left(\widehat{H}_j^{-\frac{\theta}{\sigma}} - 1\right) + \sum_j \pi_{ij} \left((\hat{\tau}_{ij}^m)^{-\theta} - 1\right) \left(\widehat{H}_j^{-\frac{\theta}{\sigma}} - 1\right)}_{\text{Labor Supply}} \\
&+ \underbrace{\sum_j \pi_{ij} \left(\widehat{\text{TMA}}_j^{\delta} - 1\right) + \sum_j \pi_{ij} \left((\hat{\tau}_{ij}^m)^{-\theta} - 1\right) \left(\widehat{\text{TMA}}_j^{\delta} - 1\right)}_{\text{Labor Demand}} \\
&+ \underbrace{\sum_j \pi_{ij} \left(\widehat{\text{TMA}}_j^{\delta} - 1\right) \left(\widehat{H}_j^{-\frac{\theta}{\sigma}} - 1\right) + \sum_j \pi_{ij} \left((\hat{\tau}_{ij}^m)^{-\theta} - 1\right) \left(\widehat{\text{TMA}}_j^{\delta} - 1\right) \left(\widehat{H}_j^{-\frac{\theta}{\sigma}} - 1\right)}_{\text{Labor Supply - Labor Demand interaction}}
\end{aligned}
$$

However, these individual components can't be identified because $\widehat{H}_j$ and $\widehat{\text{TMA}}_j$ depend on each other. Denote each component conditional on the other being fixed as follows $\tilde{\widehat{H}}_j = \widehat{H}_j \mid \widehat{\text{TMA}}_j = 1$, and $\widetilde{\widehat{\text{TMA}}}_j = \widehat{\text{TMA}}_j \mid \widehat{H}_j = 1$ and note that $\tilde{\widehat{H}}_j \neq \widehat{H}_j$ and $\widetilde{\widehat{\text{TMA}}}_j \neq \widehat{\text{TMA}}_j$. Using this notation, we can identify four channels by iteratively shutting down dimensions of endogenous response.

The first channel, the "moving *to* opportunity" channel, captures the fact that road building causes migration costs to fall. We can identify this by shutting down both the endogenous migration and trade responses by setting $\widehat{H}_j = \widehat{\text{TMA}}_j = 1$, and so identifying $\text{M2O}_i = \sum_j \pi_{ij}\left((\hat{\tau}^m_{ij})^{-\theta} - 1\right)$. Falling migration costs mean that people educated in a given location can now, on average, move to a higher-opportunity location in adulthood. This increases the opportunities available in every origin location. This effect captures the "moving *to* opportunity" channel and will result in an increase in the population in existing high-opportunity locations. This is the effect captured by reduced-form analysis that does not account for any general equilibrium effects and can be considered the direct effect of road building.

The second channel, the "moving opportunity" channel, captures the fact that migration and trade endogenously respond in spatial equilibrium to falling transport costs. This can be identified from the above exercise as $\text{MO}_i = \left[\hat{e}_i^{\theta(1-\eta)} - 1\right] - \text{M2O}_i$. This channel has three components: the endogenous response of migration (labor supply) and the endogenous response of trade (labor demand), as well as the interaction between both of these forces. Although I can't identify each of these components, I can identify the following objects.

Firstly, I can identify the *direct* labour supply effect, can be identified from setting $\widehat{\text{TMA}}_j = 1$, and calculating $\text{DirectLS}_i = \sum_j \pi_{ij}((\widetilde{\widehat{H}}_j)^{-\frac{\theta}{\sigma}} - 1) + \sum_j \pi_{ij}((\hat{\tau}^m_{ij})^{-\theta} - 1)((\widetilde{\widehat{H}}_j)^{-\frac{\theta}{\sigma}} - 1)$. By endogenising labor supply, I allow local opportunity to endogenously adjust to induced migration flows. Intuitively, higher inflows correspond to a local labour supply shock, which will crowd out opportunities by putting downward pressure on wages. This will cause wages to decrease in those areas where inflows are largest, which will be exactly those areas with preexisting higher opportunity. Conversely, local wages will rise in areas with preexisting low opportunities. However, the highest-value location is the only relevant location (conditional on individual location-specific skill shocks) when choosing where to migrate, so although this force may appear to be equilibrating, it will necessarily reduce opportunities available from each origin location. When combined with the direct effect, these two channels give the impact of road building on opportunities in frameworks that allow endogenous labour supply to impact local outcomes. The combined impact will be much more negative than the direct effect, and will normally imply that road building decreased opportunity in a given location.

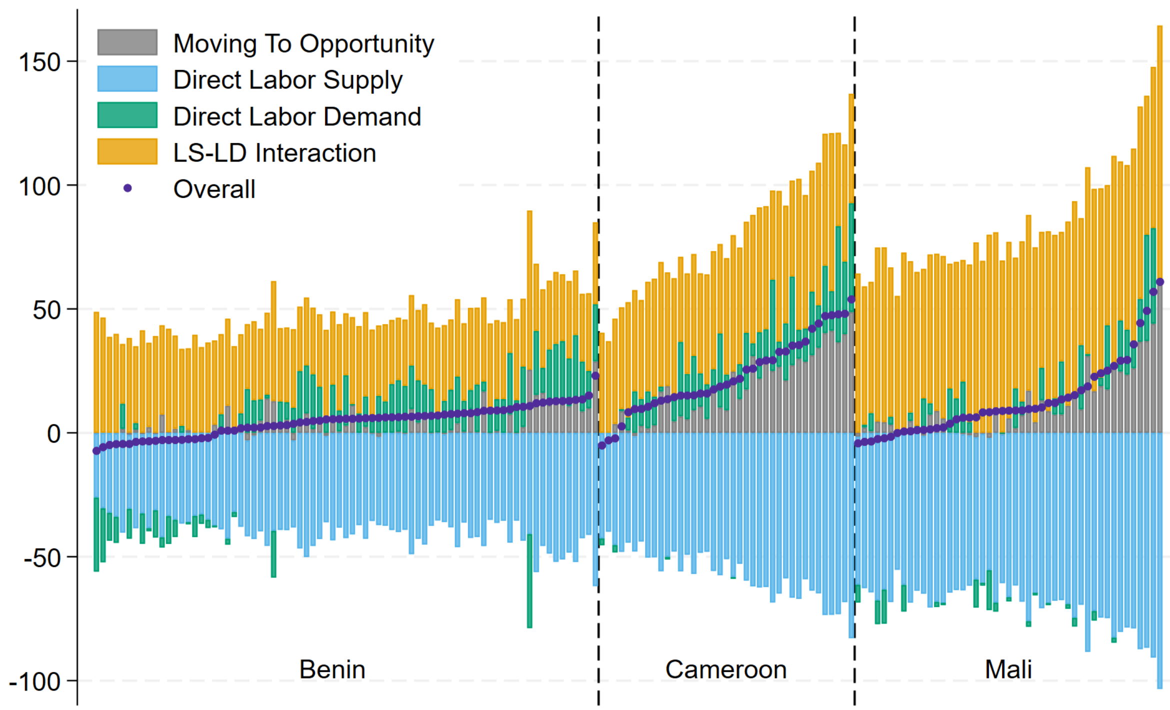
Secondly, I can identify the *direct* labor demand effect by taking the analogous approach of setting $\widehat{H}_j = 1$, and calculating $\text{DirectLD}_i = \sum_j \pi_{ij}((\widetilde{\widehat{\text{TMA}}}_j)^{\delta} - 1) + \sum_j \pi_{ij}((\hat{\tau}^m_{ij})^{-\theta} - 1)((\widetilde{\widehat{\text{TMA}}}_j)^{\delta} - 1)$. By endogenising labor demand, I allow local opportunity to adjust endogenously to changes in demand for labour caused by changes in the spatial distribution of economic activity. Reductions in transport costs cause trade costs to fall, altering spatial patterns of trade. On average, as costs have fallen, trade volumes will increase, causing wages to rise. However, prices will also adjust, and trade may be diverted from some locations that see a relatively shallower decline in transport costs costs — therefore, some locations may lose from this effect.

Thirdly, I can identify the labor supply-labor demand interaction effect that is only present when both margins are allowed to adjust endogenously. This effect will capture both the interaction term in the decomposition of the overall effects, and how the differences between $\widehat{H}_j$ and $\widetilde{\widehat{H}}_j$, and $\widehat{\text{TMA}}_j$ and $\widetilde{\widehat{\text{TMA}}}_j$, propagate through each channel. This effect can be identified as $\text{Int}_i = \text{MO}_i - \text{DirectLS}_i - \text{DirectLD}_i$. This effect captures the potentially complex interaction between endogenous migration and trade. In general, locations that received more population inflows will now see labour demand, and so wages rise. For this reason we expect the interaction effect to be positive. Note that, in the case of autarky the labour demand and labour supply effects will directly cancel, and the overall impact of roads will be equal to the direct effect.[11] Not allowing labour demand to endogenously respond would likely cause researchers to significantly underestimate the positive impact of road building on local opportunities, both through the direct and interaction effects.

Figure 5 decomposes the total effect, denoted as purple markers, into each of these channels. In gray, I plot the direct effect of decreasing migration costs and allowing more moves to opportunity. With one or two exceptions (as one or two small sections of road degenerated), this impact is positive as it effectively allows a larger location choice set for individuals to decide between in each origin location. In blue, I then isolate the direct labour supply effect. Inline with our intuition, this is resoundingly negative as the previously high-opportunity locations see their wages fall. In green I isolate the direct labour demand effect. This is, in the main, positive, but it will induce a reshuffling of demand that causes some locations to be negatively affected. Finally, in orange, I show the supply-demand interaction effect that is only present in models that allow both margins to adjust. This is also positive, as areas that saw larger increases in labour supply see the largest increases in demand. Figure 5 shows that at the locality level, not accounting for endogenous responses would result in incorrect estimates of the impact of roads, and only accounting for endogenous labour supply or labour demand individually would also result in biased average estimates. However, the degree of bias varies across countries; in Cameroon the direct effect is a fairly good approximation of the overall impact, whereas in Benin and Mali it is less so.

[11] To see this note that $\widehat{\text{TMA}}_i = \sum_j \rho_{ij}(\hat{\tau}^t_{ij})^{1-\sigma}\widehat{\text{TMA}}_j^{\frac{1-\sigma}{\sigma}}\widehat{H}_j^{\frac{\sigma-1}{\sigma}}$, and therefore as under trade-Autarky $\rho_{ii} = 1,\ \rho_{ij} = 0\ \forall j$ this implies that $\widehat{\text{TMA}}_i = \widehat{H}_i^{\frac{\sigma-1}{2\sigma-1}}$. Substituting this into our expression for market access, I find that $\widehat{\text{MA}}_i = \sum_j \pi_{ij}(\hat{\tau}^m_{ij})^{-\theta}\left(\widehat{H}_j^{\frac{\sigma-1}{2\sigma-1}}\right)^{\delta}\widehat{H}_j^{\frac{-\theta}{\sigma}} = \sum_j \pi_{ij}(\hat{\tau}^m_{ij})^{-\theta}$.

Figure 5 Mechanisms decomposition

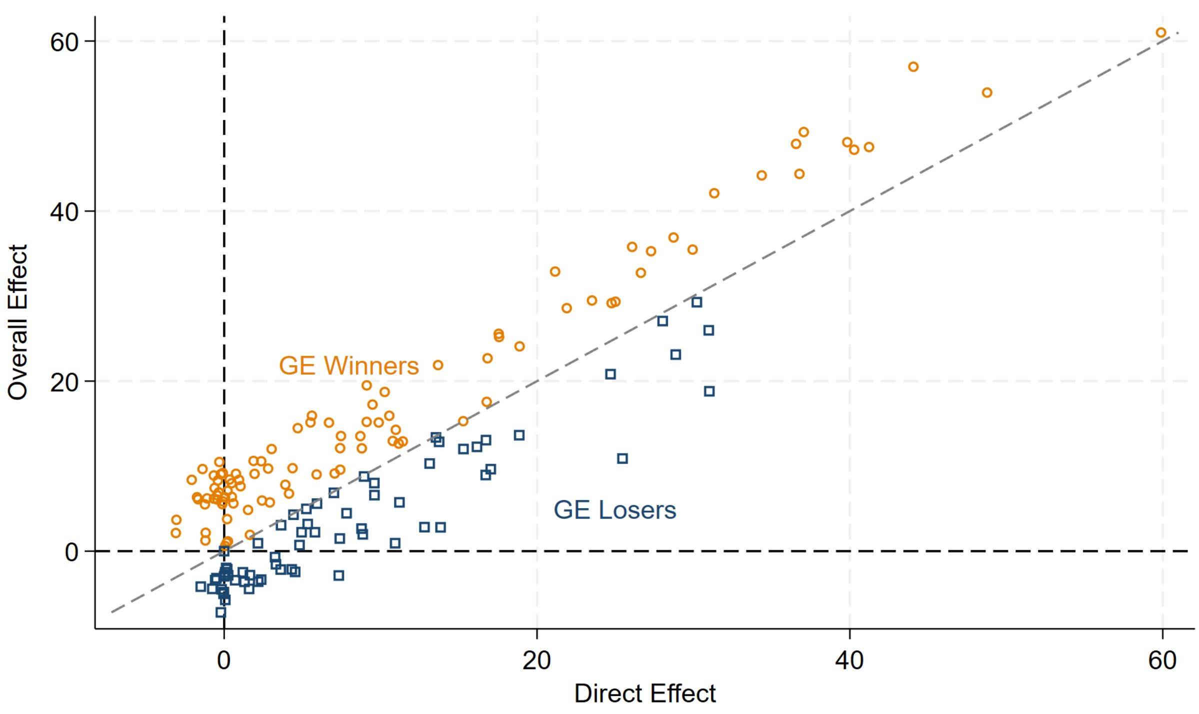


Notes: This figure decomposes the full-equilibrium location-level effect of road building since 1970 on opportunity into the constituent components of the market-access decomposition. For each location, the purple marker reports the total effect from the full model (with endogenous migration and trade). Gray bars report the direct effect of lower migration costs computed by shutting down endogenous migration and trade responses (setting $\widehat{H}_i = 1$ and $\widehat{\mathrm{TMA}}_i = 1$). Blue bars report the direct labour-supply channel computed by allowing endogenous migration but holding trade fixed (setting $\widehat{\mathrm{TMA}}_i = 1$) and taking away from this the direct effect. Green bars report the direct labour-demand channel computed by allowing trade to adjust while holding migration fixed (setting $\widehat{H}_i = 1$) and taking away the direct effect. Orange bars report the interaction term that arises only when both margins adjust, defined as the residual required for the sum of components to equal the full-model total. All effects are expressed as percent changes in opportunity.

Figure 5 suggests that only endogenising labor supply would result in significantly incorrect estimates of the aggregate impact of roads. Figure 2 in the online appendix shows this explicitly by plotting the distribution of effects across locations for three models. First, including neither endogenous margin and therefore just capturing the direct effect of moving to opportunity. Second, only endogenising labor supply. And finally, including both endogneous margins, as in my main model. The direct effect calculated from solving the model with labour supply and demand responses both shut down (mean 9.5) is almost completely positive with a long right tail. Solving the model, allowing only for the endogenous response of labour supply (mean -52.7) shifts the overall distribution significantly to the left. Solving the model, allowing for both the endogenous response of labour supply and demand (11.5) shifts the distribution back to the right, slightly further than the direct effect. The isolated aggregate endogenous response of labour supply and demand (mean 2.0) is therefore slightly positive, with a wide spread; the 10th percentile is -5.8, and the 90th is 9.3. This highlights that although most (83%) of the average effect of road building since 1970 is captured by the naive direct impact, spatial equilibrium effects can have a considerable impact on individual locations.

Figure 5 hints at winners and losers from the general equilibrium adjustments, which figure 6 spells out. On average, endogenising labour supply and demand increases the effect of road building since 1970 on opportunity by 2.0 percentage points or 21%. Figure 6 shows that winners and losers are distributed across the support of direct effects. The bite of the general equilibrium adjustments is relatively larger in locations that saw little direct impact. Opportunity in these locations could increase by as much as 10%, or decrease by similar margins, due to road building since 1970, despite the fact that travel times to and from these locations are almost completely unaffected. These often large and heterogeneous effects highlight that moving opportunity (as opposed to purely moving to opportunity) is a quantitatively relevant force in this setting.

Figure 6 Winners and losers from GE



Notes: This figure characterizes "winners" (in orange circles) and "losers" (in blue squares) from general-equilibrium adjustments by contrasting the direct migration-cost effect with the net endogenous response. The direct effect is the partial-equilibrium impact of road building since 1970 on opportunity obtained by shutting down endogenous migration and trade (setting $\widehat{H}_i = 1$ and $\widehat{\text{TMA}}_i = 1$). The net general-equilibrium adjustment is defined as the sum of the endogenous labour-supply and labour-demand responses computed as the difference between the total effect in the full model and the direct effect.

Tables 4, 5, and 6 explore heterogeneity in effects in more detail; they detail heterogeneity across each component to initial market access, initial population, and the size of the direct effect, respectively. Each table shows heterogeneity across the following components in corresponding columns: (1) the overall effect with endogenous labour supply and labour demand, (2) isolated direct effect, (3) isolated direct labour supply effect, (4) isolated ciredt labour demand effect, (5) the interaction between the endogenous labour supply and labour demand effects, and finally (6) the overall general equilibrium effects. In row one each table shows the aggregate results over all countries (including country fixed effects with robust standard errors), and in rows two, three, and four, each table shows the effects for each country individually.

Table 4 shows heterogeneity with respect to (log) initial market access, measured in the first year in which population data is available. Overall, in column one, I find that locations with higher initial market access saw larger increases in opportunity due to road building since 1970, although this effect is not statistically significant in Mali. This average positive impact masks heterogeneity by component. The direct impact doesn't systematically vary by initial market access, whereas the endogenous response of labour supply is significantly more negative in areas with preexisting high market access. This is intuitive; high market access areas are likely to be those with higher real wages and therefore those that attract an influx of people when migration costs fall, which in turn pushes down wages. Oppositely, the direct labour demand effect is increasing in initial market access, although this is only true in Benin and Cameroon. Column six shows the impact of log initial market access on the overall GE effect, which is positive.

Table 4 Heterogeneity by initial market access

	(1) Overall	(2) Direct	(3) Labor Supply	(4) Labor Demand	(5) LS-LD interaction	(6) GE Overall
Log Market Access	12.11***	2.648	-7.223***	19.54***	-2.854	9.459***
	(2.626)	(3.166)	(1.937)	(2.766)	(2.866)	(1.096)
Benin	10.10***	-1.362	-5.999***	27.80***	-10.34***	11.46***
	(1.312)	(2.107)	(1.591)	(2.598)	(1.988)	(1.245)
Cameroon	21.14***	8.133	-12.37**	30.65***	-5.270	13.01***
	(7.717)	(8.076)	(5.009)	(5.156)	(3.932)	(2.031)
Mali	11.78	7.807	-7.190	-1.488	12.65**	3.968*
	(7.751)	(8.199)	(5.415)	(4.381)	(5.089)	(2.030)
Observations	163	163	163	163	163	163

Notes: This table reports heterogeneity in the model-implied percentage change in opportunity due to road building since 1970 as a function of initial market access (measured in the first year for which population data are available in each country). Each column corresponds to a component of the decomposition: (1) the total full-equilibrium effect; (2) the direct effect; (3) the labour-supply channel net of the direct effect; (4) the labour-demand channel net of the direct effect; (5) the interaction between endogenous labour supply and labour demand, net of the direct effect and each channel in isolation; and (6) the total general-equilibrium adjustment. Row 1 pools all countries and includes country fixed effects. Subsequent rows report country-specific estimates. The unit of observation is a location; all locations are weighted equally, and robust standard errors are reported in parentheses.

Table 5 then shows heterogeneity in the impact of road building since 1970 on opportunity by initial population, again measured in the first available year in each country. Column one shows that overall areas with higher population saw larger gains, and this is true in all three countries, although not statistically significant in Cameroon. This aggregate positive effect is, however, driven by different channels in each setting. In Benin, areas with a higher initial population saw a larger labour demand effect, mitigated but not nullified by the endogenous migration reaction of individuals. In Cameroon, the positive reaction comes mainly from the direct effect; areas with higher initial population, are those areas where migration costs to higher-opportunity locations fell the most. Finally, in Mali, the correlation is driven by the direct effect as in Cameroon. However, in Mali, this direct effect is mitigated by the endogenous migration response of individuals, which itself is attenuated by the reaction of firms.

Table 5 Heterogeneity by initial population

	(1) Overall	(2) Direct	(3) Labor Supply	(4) Labor Demand	(5) LS-LD interaction	(6) GE Overall
Log Population	5.905***	5.713***	-4.236***	2.431	1.996	0.191
	(2.250)	(2.017)	(1.483)	(1.824)	(1.580)	(0.870)
Benin	3.866***	1.227	-1.972	8.140**	-3.529**	2.639**
	(1.354)	(1.586)	(1.576)	(3.174)	(1.366)	(1.328)
Cameroon	2.950	4.568	-1.994	0.333	0.0415	-1.619
	(3.949)	(3.178)	(2.220)	(2.905)	(1.524)	(1.559)
Mali	11.55**	11.82***	-9.280***	-1.146	10.16***	-0.268
	(4.516)	(4.161)	(2.755)	(2.825)	(2.701)	(1.324)
Observations	163	163	163	163	163	163

Notes: This table reports heterogeneity in the model-implied percentage change in opportunity due to road building since 1970 as a function of initial population (measured in the first year for which population data are available in each country). Each column corresponds to a component of the decomposition: (1) the total full-equilibrium effect; (2) the direct effect; (3) the labour-supply channel net of the direct effect; (4) the labour-demand channel net of the direct effect; (5) the interaction between endogenous labour supply and labour demand, net of the direct effect and each channel in isolation; and (6) the total general-equilibrium adjustment. Row 1 pools all countries and includes country fixed effects. Subsequent rows report country-specific estimates. The unit of observation is a location j; all locations are weighted equally, and robust standard errors are reported in parentheses.

Finally, in table 6, I consider how the overall effect and each component are correlated with the direct effect. Unsurprisingly, column one shows a positive correlation on average and across countries. By comparing coefficients across countries, we can see that the direct effects is very predictive of the overall impact on average in Cameroon and Mali, but much less so in Benin. Column two shows that areas with the largest direct effect saw the largest negative impacts due to the endogenous reaction of labour supply — this is intuitive as the most positively affected areas see the largest in-migration and therefore negative impact on local wages. Analagously, the labour demand effect is more positive in these areas as their better connectedness brings in more trade demand.

Table 6 Heterogeneity by direct effect

	(1) Overall	(2) Labor Supply	(3) Labor Demand	(4) LS-LD interaction	(5) GE Overall
Direct Effect	0.963***	-0.702***	0.366***	0.299***	-0.0373
	(0.0478)	(0.0270)	(0.119)	(0.0846)	(0.0478)
Benin	0.534***	-0.755***	0.173	0.115	-0.466***
	(0.0894)	(0.129)	(0.411)	(0.226)	(0.0894)
Cameroon	1.029***	-0.677***	0.440***	0.267***	0.0293
	(0.0595)	(0.0404)	(0.126)	(0.0826)	(0.0595)
Mali	1.055***	-0.705***	0.371**	0.390***	0.0553
	(0.0718)	(0.0240)	(0.187)	(0.143)	(0.0718)
Observations	163	163	163	163	163

Notes: This table reports heterogeneity in the model-implied percentage change in opportunity due to road building since 1970 as a function of the direct effect of road building calculated shutting down the endogenous labour supply and demand responses. Each column corresponds to a component of the decomposition: (1) the total full-equilibrium effect; (2) the labour-supply channel net of the direct effect; (3) the labour-demand channel net of the direct effect; (4) the interaction between endogenous labour supply and labour demand, net of the direct effect and each channel in isolation; and (5) the total general-equilibrium adjustment. Row 1 pools all countries and includes country fixed effects. Subsequent rows report country-specific estimates. The unit of observation is a location; all locations are weighted equally, and robust standard errors are reported in parentheses.

Finally, figures 3, 4, and 5 in the appendix show the spatial distribution of the impact of road building since 1970 on maps in Benin, Cameroon, and Mali. In each country these maps tell the same intuitive story. Locations that became better connected see the largest direct effects, the most negative direct labour supply effect, and the most positive labour demand and labor supply - labor demand interaction effects.

6 Conclusion

Opportunity is unevenly distributed across space. This paper asks the policy-relevant question: Can investments in connectivity change this geography of opportunity, defined as the causal effect of place on primary education completion? Focusing on Benin, Cameroon, and Mali, I find that road building does alter local opportunity both by encouraging moves to opportunity, and by altering the underlying distribution of opportunity itself.

Measuring the causal effect of changes in road building is complicated by spillovers across space and the non-random placement of connections. To overcome these challenges, I first turn to theory combining a spatial economics model with education choice and endogenous real wages with a place-effects framework. This emits a sufficient statistic result, relating primary completion to local market access that can be taken to the data. To overcome the endogeneity of road placement, I develop a novel not-on-least-cost-path identification strategy that isolates indirect network-mediated variation in market access. Finally, to identify causal place effects from selection effects, I use a movers design, leveraging variation in exposure due to age at move. I find that growing up in a one standard deviation higher market access location increases the probability of completing primary school by about 7 percentage points, or roughly 12%.

To translate these reduced-form effects into the aggregate implications of historical road building, I solve and credibly estimate the model. I find that road building since 1970 increased local opportunity by on average 11.%, but that this average hides considerable heterogeneity, with 17% of locations experiencing negative impacts. Considering mechanisms, I find that although most of the average impact can be explained by the direct effect of easing moves to opportunity, endogenous labour supply and demand responses are crucial to understanding location-specific heterogeneity in effects.

Three implications follow. First, connectivity investments can increase local opportunities. Second, distributional consequences are central. Changes to the road network can create local winners and losers beyond the direct changes in connectivity they induce because improvements in one corridor affect the entire system through the endogenous adjustment of labour supply and demand. Third, as a result of this endogenous reshuffling, there can be absolute opportunity losers from road building — it is far from a silver bullet.

References

Adamopoulos, T. (2025). Spatial integration and agricultural productivity: Quantifying the impact of new roads. *American Economic Journal: Macroeconomics 17*(1), 343–378.

Adukia, A., S. Asher, and P. Novosad (2020). Educational investment responses to economic opportunity: evidence from indian road construction. *American Economic Journal: Applied Economics 12*(1), 348–76.

Alesina, A., S. Hohmann, S. Michalopoulos, and E. Papaioannou (2021). Intergenerational mobility in africa. *Econometrica 89*(1), 1–35.

Allen, T. and C. Arkolakis (2023). Economic activity across space: a supply and demand approach. *Journal of Economic Perspectives 37*(2), 3–28.

Allen, T., C. Arkolakis, and X. Li (2024). On the equilibrium properties of spatial models. *American Economic Review: Insights 6*(4), 472–489.

Allen, T., C. Arkolakis, and Y. Takahashi (2020). Universal gravity. *Journal of Political Economy 128*(2), 393–433.

Alpern, S. B. (1999). Dahomey's royal road. *History in Africa 26*, 11–24.

Asher, S. and P. Novosad (2020). Rural roads and local economic development. *American economic review 110*(3), 797–823.

Association, I. D. (1981). Mali – road maintenance project: Credit 1104 – credit agreement – conformed. World Bank Documents & Reports. Credit No. 1104 MLI (conformed copy).

Atkin, D. (2016). Endogenous skill acquisition and export manufacturing in mexico. *American Economic Review 106*(8), 2046–85.

Banerjee, A., E. Duflo, and N. Qian (2020). On the road: Access to transportation infrastructure and economic growth in china. *Journal of Development Economics 145*, 102442.

Bank, W. (1970, sep). Dahomey – highway maintenance and engineering project: Credit 0215 – credit agreement – conformed. World Bank Documents & Reports.

Bank, W. (1982). Project performance audit report: Cameroon second and third highway projects. Evaluation of the Cameroon Second Highway Project and Third Highway Project; covers Loan 935-CM, Credit 429-CM (including supplementary financing), and Loan 1515-CM.

Baum-Snow, N., L. Brandt, J. V. Henderson, M. A. Turner, and Q. Zhang (2017). Roads, railroads, and decentralization of chinese cities. *Review of Economics and Statistics 99*(3), 435–448.

Baum-Snow, N., J. V. Henderson, M. A. Turner, Q. Zhang, and L. Brandt (2020). Does investment in national highways help or hurt hinterland city growth? *Journal of Urban Economics 115*, 103124.

Borusyak, K. and P. Hull (2023). Nonrandom exposure to exogenous shocks. *Econometrica 91*(6), 2155–2185.

Brooks, W. and K. Donovan (2020). Eliminating uncertainty in market access: The impact of new bridges in rural nicaragua. *Econometrica 88*(5), 1965–1997.

Bryan, G., S. Chowdhury, and A. M. Mobarak (2014). Underinvestment in a profitable technology: The case of seasonal migration in bangladesh. *Econometrica 82*(5), 1671–1748.

Bryan, G. and M. Morten (2019). The aggregate productivity effects of internal migration: Evidence from indonesia. *Journal of Political Economy 127*(5), 2229–2268.

Burgess, R., R. Jedwab, E. Miguel, A. Morjaria, and G. Padró i Miquel (2015). The value of democracy: evidence from road building in kenya. *American Economic Review 105*(6), 1817–51.

Castaing Gachassin, M. (2013). Should i stay or should i go? the role of roads in migration decisions. *Journal of African Economies 22*(5), 796–826.

Chetty, R. and N. Hendren (2018a). The impacts of neighborhoods on intergenerational mobility i: Childhood exposure effects. *The Quarterly Journal of Economics 133*(3), 1107–1162.

Chetty, R. and N. Hendren (2018b). The impacts of neighborhoods on intergenerational mobility ii: County-level estimates. *The Quarterly Journal of Economics 133*(3), 1163–1228.

Chetty, R., N. Hendren, and L. F. Katz (2016). The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment. *American Economic Review 106*(4), 855–902.

Chiovelli, G., S. Michalopoulos, E. Papaioannou, and S. Sequeira (2025). Civil war-induced displacement and human capital. Technical report, Facultad de Ciencias Empresariales y Economia. Universidad de Montevideo.

Chyn, E. and D. Daruich (2025). An equilibrium analysis of the effects of neighborhood-based interventions on children. *American Economic Review 115*(12), 4476–4522.

Deutscher, N. (2020). Place, peers, and the teenage years: Long-run neighborhood effects in australia. *American Economic Journal: Applied Economics 12*(2), 220–49.

Donaldson, D. (2018). Railroads of the raj: Estimating the impact of transportation infrastructure. *American Economic Review 108*(4-5), 899–934.

Donaldson, D. (2022). Blending theory and data: A space odyssey. *Journal of Economic Perspectives 36*(3), 185–210.

Donaldson, D. and R. Hornbeck (2016). Railroads and american economic growth: A "market access" approach. *The Quarterly Journal of Economics 131*(2), 799–858.

Eckert, F. and T. Kleineberg (2024). The geography of opportunity: Education, work, and intergenerational mobility across us counties. Technical report, Technical report, Working Paper.

Edmonds, E. V., N. Pavcnik, and P. Topalova (2010). Trade adjustment and human capital investments: Evidence from indian tariff reform. *American Economic Journal: Applied Economics 2*(4), 42–75.

Faber, B. (2014). Trade integration, market size, and industrialization: evidence from china's national trunk highway system. *Review of Economic Studies 81*(3), 1046–1070.

Foster, V. and C. Briceño-Garmendia (2010). *Africa's infrastructure: a time for transformation*. Agence Française de Développement.

Fujimoto, J., D. Lagakos, and M. VanVuren (2023). Aggregate and distributional effects of 'free' secondary schooling in the developing world.

Ghani, E., A. G. Goswami, and W. R. Kerr (2016). Highway to success: The impact of the golden quadrilateral project for the location and performance of indian manufacturing. *The Economic Journal 126*(591), 317–357.

Graff, T. (2024). Spatial inefficiencies in africa's trade network. *Journal of Development Economics 171*, 103319.

Gwilliam, K. (2011). *Africa's transport infrastructure: Mainstreaming maintenance and management*. The World Bank.

Hsiao, A. (2024). Educational investment in spatial equilibrium: Evidence from indonesia.

IPUMS (2020). Integrated public use microdata series, international: Version 7.2 [dataset].

Jedwab, R., E. Kerby, and A. Moradi (2017). History, path dependence and development: Evidence from colonial railways, settlers and cities in kenya. *The Economic Journal 127*(603), 1467–1494.

Jedwab, R. and A. Moradi (2016). The permanent effects of transportation revolutions in poor countries: evidence from africa. *Review of economics and statistics 98*(2), 268–284.

Jedwab, R. and A. Storeygard (2021). The average and heterogeneous effects of transportation investments: Evidence from sub-saharan africa 1960-2010. *Journal of the European Economic Association*.

Kebede, H. A. (2024). Gains from market integration: Welfare effects of new rural roads in ethiopia. *Journal of Development Economics 168*, 103252.

Khanna, G. (2022). Large-scale education reform in general equilibrium: Regression discontinuity evidence from india. *The American Economic Review*.

Laliberté, J.-W. P. (2021). Long-term contextual effects in education: Schools and neighborhoods. *American Economic Journal: Economic Policy*.

Michaels, G. (2008). The effect of trade on the demand for skill: Evidence from the interstate highway system. *The Review of Economics and Statistics 90*(4), 683–701.

Milsom, L. H. (2023). Spatial inequality of opportunity in west africa. *Journal of Public Economics 225*, 104944.

Moneke, N. (2020). Can big push infrastructure unlock development? evidence from ethiopia. Technical report, Working paper.

Morten, M. and J. Oliveira (2024). The effects of roads on trade and migration: Evidence from a planned capital city. *American Economic Journal: Applied Economics 16*(2), 389–421.

Mukherjee, M. (2012). Do better roads increase school enrollment? evidence from a unique road policy in india.

Okoye, D., R. Pongou, and T. Yokossi (2019). New technology, better economy? the heterogeneous impact of colonial railroads in nigeria. *Journal of Development Economics 140*, 320–354.

Psacharopoulos, G. and H. A. Patrinos (2018). Returns to investment in education: a decennial review of the global literature. *Education Economics 26*(5), 445–458.

Redding, S. J. and E. Rossi-Hansberg (2017). Quantitative spatial economics. *Annual Review of Economics 9*, 21–58.

Redding, S. J. and M. A. Turner (2015). Transportation costs and the spatial organization of economic activity. *Handbook of regional and urban economics 5*, 1339–1398.

Rojas Ampuero, F. C. (2022). *Sent Away: The Long-Term Effects of Slum Clearance on Children and Families*. Ph. D. thesis, UCLA.

Silva, J. S. and S. Tenreyro (2006). The log of gravity. *The Review of Economics and statistics 88*(4), 641–658.

Simonovska, I. and M. E. Waugh (2014). The elasticity of trade: Estimates and evidence. *Journal of international Economics 92*(1), 34–50.

Sotelo, S. (2020). Domestic trade frictions and agriculture. *Journal of Political Economy 128*(7), 2690–2738.

Tchindjang, M., E. David, A. Bopda, J. Tanga Onana, Z. Nji Fogwe, and C. Mbifung Lambi (2005). Mapping of main artery network and colonial porterage in cameroon (1900–1940). In *ICC 2005: Proceedings of the 22nd International Cartographic Conference — Mapping Approaches into a Changing World*, A Coruña, Spain. International Cartographic Association (ICA).

UN (2018). 2018 revision of world urbanization prospects.

Van Maarseveen, R. (2025). Urbanization and educational attainment: Evidence from africa. *Journal of Development Economics*, 103654.

Weidmann, N. B., J. K. Rød, and L.-E. Cederman (2010). Representing ethnic groups in space: A new dataset. *Journal of Peace Research 47*(4), 491–499.

World Bank (1994, March). Republic of benin: Transport sector strategy. Report 12098-BEN, World Bank. Confidential; prepared jointly by the Ministry of Public Works and Transport (MTPT) and the World Bank.

Yotov, Y. V., R. Piermartini, J.-A. Monteiro, and M. Larch (2016). *An advanced guide to trade policy analysis: The structural gravity model*. World Trade Organization Geneva.

Zant, W. (2022). Measuring trade cost reductions through a new bridge in mozambique: Who benefits from transport infrastructure? *Journal of African Economies 31*(4), 384–408.